

**BACK TO
METAL**

WHY THIS BOOK EXISTS

YOU ARE ALLOWED TO RUN YOUR OWN COMPUTERS

A generation of engineers has now been trained to believe that owning a server is a kind of professional failure. It is not. It is a trade, and like every trade it has terms: you choose dedicated capacity, rented or owned in colocation, and keep more of the budget as your workload grows. Existing cloud ops engineers transition to on-premises platform work; contracted physical tasks go to remote hands, already priced in the facility bill, or to the rental provider. The model keeps the same hours and salary. On-premises roles can cost less than cloud specialist roles, but those potential savings are not counted. For a company with steady demand and a substantial cloud bill, both routes deserve a serious comparison.

What has been missing is not the argument. It is a plan short enough to finish. Repatriation is written about as a decision and executed as a crisis, which is why so many attempts stall halfway, with two platforms live and nobody able to say whether it is going well. This is the other thing: 20 Moves, in five stages, each one a single job with an estimated cutover, a stated risk and explicit limits on reversal. An earlier edition of this book had a hundred and twenty-two of them and took four years. Nobody has four years.

It is honest about the parts that do not pay. One whole Move ends by telling you to keep paying somebody else, because a content network, denial-of-service scrubbing and outbound email are three things you will not beat on your own, and a book that pretends otherwise is selling something. Another gives you permission to read three Moves, do the arithmetic, and stop. The point was never to own everything. It was to own the parts where ownership is cheaper, faster and yours.

Which is why the whole thing is **open source**. Every Move, the typesetter that turns them into this book, and the site that publishes them are yours: the text under Creative Commons, the software under the MIT licence, all of it at github.com/OneUptime/back-to-metal. Correct it, extend it, add the services your own estate actually runs. Infrastructure knowledge this practical should not sit behind a consulting invoice.

BACK TO METAL

How a company leaves the cloud, one move at a time

Nawaz Dhanda

BY THE MAKERS OF ONEUPTIME • [ONEUPTIME.COM](https://oneuptime.com)

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Every runbook in this book changes production infrastructure, and several of them move data that cannot be recreated. Read the Rollback section before the first step of any Move, take a backup you have restored at least once, and rehearse against a copy. Prices, service names and product behaviour are those the author observed while writing and will drift; the figures in each Move are illustrative of the shape of a saving, not a quotation. Nothing here is legal, financial, security or compliance advice. You remain responsible for your own systems, your own data and your own obligations to the people whose data it is.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Move would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are licensed under the SIL Open Font License.

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MANY MOVES, NOT ONE MIGRATION

The reason leaving the cloud fails is almost never that the technology did not work. It is that it was attempted as one decision, executed as one project, and abandoned somewhere in the middle with two platforms running and nobody able to say whether it was going well.

This book takes the opposite shape. It is 20 Moves, in five stages. Each is one job with a stated cutover, a stated risk and a rollback that has been thought about. Rehearse the return path before moving traffic or data. Some commitments and deletions cannot be undone; each Move states the conditions for stopping safely.

It is deliberately small. The first edition had a hundred and twenty-two Moves and, computed from its own files, 975 person-days and four and a half years. That was correct and useless. This is 107 person-days and about 41 weeks for two people. Procurement and observation windows extend the calendar without consuming engineer-days; extra people cannot remove those waits.

They are ordered so that a reader who starts at Move 01 and works forward has, by construction, met every prerequisite of the Move in front of them. That is not a stylistic choice; the build refuses to compile a book where it does not hold.

Every Move carries all three clouds. The job is the same whether you are leaving AWS, Google Cloud or Azure — what differs is the extraction, so each Move opens with three lines naming the real service on each provider and the one thing that is different there.

Move 04 concludes that you should keep paying somebody else for three things. A content delivery network, scrubbing capacity at the edge and outbound email are businesses other people run better than you will, and each is cheap next to what it replaces. The aim was never to own everything. And Move 03 gives you permission to stop: if the arithmetic does not clear a third, stop after the invoice, inventory and measurement work.

Rehearse everything, and read the Rollback before the first step of every Move — including the ones that look like nothing.



EVERY MOVE, WITHOUT EXCEPTION

HOW A MOVE WORKS

- 1 THE RING**
The cutover: how many minutes of user-visible downtime this Move costs, against a 60-minute dial. An empty ring means none, and most rings are empty.
- 2 THE BARS**
Risk, meaning blast radius if it goes wrong rather than how fiddly it is. Three bars and you do not run it alone.
- 3 THE PANEL**
Everything that must already be true before step one. Hardware, software with versions, and the access you will be asked for.
- 4 WHY THIS WORKS**
What the managed service actually was, and why the replacement holds. Read it once; you will not need it again.
- 5 THE STRIP**
What it cost, what it costs now, the saving, the downtime and the person-days. Illustrative of a shape, not a quotation for your account.
- 6 THE ROLLBACK**
On the facing page, at the foot, in colour. It names the point of no return. Read it before step one, not after step five.

WHY LEAVE AT ALL

Colocation and rented dedicated servers put more of your infrastructure budget into capacity you can use. Rent the machines to preserve cash, or own them in a colocation facility for lower running costs and control over the hardware. In this reference

model, colocation saves \$4,886,320 over five years, with the same operations hours as the cloud. Your team brings its skills; the provider supplies the building and physical support.

THE EVIDENCE TO COLLECT BEFORE CLAIMING A WIN

Uptime

Measure from outside both estates, with the same service boundary and observation window. Count site failures and planned downtime.

Latency

Replay representative traffic on both platforms. Compare throughput and tail latency at the same offered load.

Repairs

Record physical interventions, time to recovery and remote-hands charges. A response-time promise is not a repair-time measurement.

Hours

Record recurring platform work on both sides. Separate migration labour from steady-state operations and on-call coverage.

The financial figures are a worked model. They do not establish an availability, performance or staffing result for your estate.

1 THE BILL STOPS BEING A PERCENTAGE OF YOUR GROWTH

This is the one that matters and the rest are consequences of it. A cloud bill is a toll on activity: more users, more bytes, more bill, for ever, at a margin somebody else sets and can change. A rack or a dedicated-server contract buys capacity at a predictable price. While your workload fits that capacity and your power and traffic allowances, more users need not mean more infrastructure rent. Add machines when demand calls for them. Over five years the difference here is \$4,886,320, and the machines are still working at the end of it.

CHECK IT

Take last quarter's invoices and plot the total against your own usage metric. If the line is flat you have nothing to gain here. If it tracks your growth, that slope is what you are buying out of — Move 01.

2 EGRESS STOPS BEING A TAX ON YOUR OWN TRAFFIC

Moving your own data to your own users is where the margin lives, and it is the line to price explicitly: \$7,980 a month for 100 TB of AWS internet egress at the model's US East list rates. Facility transit replaces that meter with a bandwidth commitment and possible overage charges. Compare the actual traffic pattern and contract on both sides.

CHECK IT

Find the data-transfer line on last month's bill. Divide it by your egress in terabytes. Compare that effective rate with a transit quote and the edge services you retain — Move 01, then Move 18.

3 THE MACHINES ARE YOURS, SO THE PERFORMANCE IS YOURS

A vCPU can be an SMT thread or a whole physical core; the instance family decides. Dedicated hardware gives you control over placement and contention, and local NVMe changes the storage path. Neither guarantees faster queries: CPU generation, memory, disks and workload still decide the result.

CHECK IT

Compare your ninety-ninth percentile against your median for a week. If the gap is wide, investigate contention and application behaviour. Replay the same load on real hardware in Move 14 before attributing the gap to the cloud.

4 YOU CONTROL THE UPGRADE WINDOW

A managed service is somebody else's roadmap running inside your product. Instance families are retired, versions go end-of-life on a date you did not pick, a control plane upgrades on its own release channel, and the longest you can defer an upgrade depends on the service and support policy. On your own platform you schedule the upgrade, but software and firmware still reach end of support. Budget patching and replacement; ownership does not make an unsupported version safe.

CHECK IT

Count the forced upgrades and deprecation notices you have absorbed in the last two years, and what each cost in engineer-days. That is a recurring bill nobody invoices you for.

5 YOUR TEAM RUNS THE PLATFORM; REMOTE HANDS LOOK AFTER THE RACK

Your cloud-ops engineers already deploy, monitor, patch and recover services. Those skills move with the workload. Colocation remote hands carry out disk swaps, cabling and power cycles under your runbooks; a rented-metal provider maintains its hardware under the support agreement. Physical support is already priced in facility charges or rent. The model retains the existing team, hours and salary, without counting potential savings from lower-cost on-premises roles.

CHECK IT

Agree the physical tasks, coverage and response times with the provider, then rehearse an incident together. Measure your team's hours before and after the move — Moves 07 and 19.

CHOOSE HOW YOU MOVE

Both routes can lower the bill. Choose the purchasing and support arrangements that fit your

cash flow and the team you already have.

1 RENT FIRST, OR OWN IN COLOCATION

Rented metal lets you move without buying the servers upfront. Colocation lets you own the hardware and spread its purchase cost over years of use. Compare both cash flows in Move 03 and choose the route in Move 07.

2 A TIMETABLE THAT FITS THE ROUTE

Available rented servers can avoid the hardware purchase and facility setup lead times. The colocation plan includes ordering machines, signing space and waiting for circuits: 107 person-days of work across 41 weeks. Both routes keep the migration rehearsals and rollback windows.

3 THE EXISTING TEAM, WITH NO ADDED SALARY IN THE MODEL

Cloud ops transitions into on-prem ops. Remote hands handles the physical interventions; your engineers keep responsibility for the platform and application. The model budgets 160 hours a month in every option, with remote hands in the facility bill. On-premises roles can cost less than cloud specialist roles; that potential saving is excluded by keeping the same salary. Validate local rates and hours. Migration work is counted separately.

4 THREE THINGS THAT NEVER COME HOME

A content delivery network, outbound mail deliverability and scrubbing at the edge are businesses other people run better than you will. Move 04 tells you to keep paying for all three. These and the services retained in later Moves total \$9,180 a month in the reference model.

WHEN TO STAY EXACTLY WHERE YOU ARE

Your load is spiky or seasonal and genuinely scales to near nothing between the spikes. Elasticity is the one thing a rack cannot do, and paying for the peak all month is how owning becomes the expensive answer.

You have no team to own the platform and no managed operations partner. Remote hands covers physical work; application recovery and platform decisions still need an owner. An existing cloud-ops team can transition into that role.

You lean hard on managed services that have no real equivalent - a serverless database, a hosted stream, a workflow engine - and replacing them is a rewrite rather than a migration. Move 02 will tell you this in an afternoon.

Your effective cloud bill is too small to cover migration and the capacity you need. Price rented metal as well as a colocation rack: renting can suit a smaller estate. Move 03 compares your own quotes and hours, so the decision rests on your workload rather than a spending threshold borrowed from another company.

If those reasons to stay do not apply and the arithmetic in Move 03 clears, the remaining Moves provide a staged plan, with a stated rollback limit for each job.

THE REFERENCE BUILD

Every Move in this book is written against one cluster, so that a runbook can name a real thing rather than a category: one site, 16 nodes and 2 on the shelf, 32 cores and 256 GB a node, 25 GbE to the switch. 16 because that is what the bill buys and a little over, and because at this size losing one costs six per cent of the fleet rather than a third; the spares because a dead board is a return authorisation and three weeks. Scale the numbers; do not scale away the redundancy.

Cloud VMs share Proxmox VE hosts as KVM guests. Containers run on Talos and Kubernetes, on hardware or in guests. Classify workloads in Move 02 before assigning hosts in Move 11. This prices a container estate at one site with off-site recovery. Add VM support, licences, backups and any required second site. Validate operating hours in Move 03. Totals exclude transition labour, overlapping cloud bills, financing and taxes.

COMPUTE

- ☐ 16 1U dual-socket servers, and 2 on the shelf
- ☐ 32 physical cores a node, current-generation EPYC or Xeon
- ☐ 256 GB ECC memory a node, in a vendor-approved balanced configuration
- ☐ Redundant hot-swap power supplies, each fed from a different strip
- ☐ A management controller on every node - Redfish, not just a web console

STORAGE

- ☐ Four NVMe drives a node, mixed-use endurance, 3.84 TB each
- ☐ Enterprise drives with power-loss protection - not consumer SSDs
- ☐ A separate boot pair, mirrored
- ☐ One spare drive of each size, on site, unopened
- ☐ A backup target at a provider you are not otherwise using

NETWORK AND SPACE

- ☐ Two 25 GbE switches, stacked or in a link-aggregation pair
- ☐ Two uplinks a node, one to each switch
- ☐ A separate 1 GbE management switch, not reachable from the internet
- ☐ A full rack in a carrier-neutral facility, A and B power
- ☐ Facility transit for go-live, and a second upstream ordered alongside it
- ☐ An out-of-band line that works when the main path does not

THE SOFTWARE STACK

- ☐ Proxmox VE on VM hosts, with KVM guests for Linux and Windows workloads
- ☐ Talos Linux for Kubernetes nodes, on hardware or in KVM guests
- ☐ Kubernetes, on a supported version compatible with the whole stack
- ☐ Cilium for the cluster network, with kube-proxy replaced
- ☐ Ceph owned by Proxmox for VM disks or by Rook on direct Talos nodes
- ☐ CloudNativePG, on local NVMe rather than on Ceph
- ☐ cert-manager before database backup plugins; Envoy Gateway at the edge
- ☐ Argo CD, with the cluster state in Git
- ☐ Prometheus and Loki for the stores, OneUptime for what wakes somebody
- ☐ Velero for Kubernetes; separate whole-VM backups, each restore rehearsed

ACCESS

- ☐ An identity provider that is not the cluster
- ☐ Break-glass credentials in a safe, on paper
- ☐ A jump host or a mesh - never a public API server
- ☐ Hardware tokens for anyone who can delete a cluster
- ☐ A status page that is not hosted on the thing it reports on
- ☐ The domain registrar account locked, with recovery codes printed

ON THE LAPTOP

- ☐ kubectl
- ☐ helm
- ☐ talosctl
- ☐ k9s
- ☐ argocd
- ☐ velero
- ☐ psql and pg_dump
- ☐ barman-cloud
- ☐ redis-cli
- ☐ rclone
- ☐ the cloud CLI, still logged in
- ☐ dig and mtr
- ☐ openssl
- ☐ ipmitool
- ☐ the runbook, printed

THE ON-RAMP

Almost nobody should sign a facility contract before they have run this stack once. A cluster of 3 refurbished machines with 12 cores and 64 GB each, on a managed switch under a desk, lets you rehearse the chosen platform and a service migration. Adapt the disk layout to the smaller kit. It costs about \$3,430 once and about \$30 a month in electricity, it saves nothing whatsoever, and it is the cheapest way to find out whether the rest of this book is for you.

WHY THREE, AND WHY SECOND-HAND

Three because a control plane needs three members to lose one and keep going, and everything you will learn about quorum, fencing and upgrades needs somewhere to go wrong. Second-hand because a machine two generations old runs Kubernetes exactly as well as a new one at roughly a tenth of the price, and because you want to be willing to break it.

WHAT IT CANNOT TEACH YOU

Every word of Stage 2. A homelab has one power feed, one switch, no cross-connect, no remote hands and nobody to escalate to at three in the morning. It cannot show you what a colocation contract is for, or what happens when the A feed goes away and the B feed was never tested. The building is a different discipline, which is why it has a stage of its own.

THE PART TO CHECK CAREFULLY

The drives. Without power-loss protection, a sudden outage can lose acknowledged writes or damage data. Use enterprise NVMe with that protection for durable state, check drive health, and rehearse recovery. A UPS helps with interruptions but does not replace the drive's guarantees or independent backups.

WHAT IT PROVES

For VMs, Proxmox guest provisioning, networking and a backup restored on another host. For containers, Talos, Kubernetes and a Postgres cutover against a copy. Rehearse the storage branch you intend to run; this small kit does not prove production capacity or recovery time.

WHEN TO STOP USING IT

Never. The on-ramp becomes the non-production cluster the rest of the book keeps asking for — the one you restore etcd onto, rehearse an upgrade on, and point a load generator at. Every estate needs a cluster it is allowed to destroy, and this is the cheapest one you will ever own.

IF THREE MACHINES AT HOME IS NOT POSSIBLE

Rent three suitable dedicated hosts available on monthly terms and run the same experiment. It costs more than electricity and less than being wrong, it gives you real addresses and a real network, and you can end the rental under the provider's notice terms. What it will not give you is a machine you can physically pull a disk out of.

THE WHOLE SHOPPING LIST

- ☐ Three refurbished mini PCs or small-form-factor desktops
- ☐ 64 GB in each - memory is what runs out first, not cores
- ☐ Two NVMe drives a node: one to boot, one for data
- ☐ A managed switch with VLANs and a spare port for out-of-band
- ☐ A small uninterruptible power supply, mostly to survive the brownouts that corrupt etcd
- ☐ A domain you own, for real certificates rather than self-signed ones

THE RULES FOR LEAVING THE CLOUD

-
- 1 **Move the cheapest thing first.** The first service cutover exists to prove the platform and the team, and it should be something whose failure is a shrug. You are buying evidence, and evidence is worth more early than savings are.
 - 2 **Data moves last and leaves last.** Stateless traffic can return quickly when both paths and client behaviour have been tested. State needs a transfer plan. Move compute first, run it against the cloud database across the link, and only then move the database - by which point you have already learned the things you would otherwise learn during a data migration.
 - 3 **Never run two platforms longer than you planned to.** The cost of leaving is not the hardware. It is the months in which both estates are live, both are on call, and every change has to be made twice. Pick the overlap window before you start, write it down, and treat overrunning it as the incident it is.
 - 4 **Keep the bill until the invoice proves you can stop.** A resource is not decommissioned when nothing points at it. It is decommissioned when a full month has passed, the line has gone from the bill, and nobody has filed a ticket. Turning things off too early is how a migration becomes an outage a fortnight later.
 - 5 **Buy the second of everything, and one spare.** Two switches, two power feeds, two people who can get into the building, and spare machines that are already burned in. Every single point of failure you accept at the start becomes an incident you attend in person, at night, in a year.
 - 6 **Count the salary.** A saving that costs engineer-hours is a real saving only if you write the hours into the comparison - on both sides, because the cloud takes them too. Most of these still win by a wide margin once you do. The ones that do not, you want to know about in week one rather than month ten.
 - 7 **Keep paying for the three things you will not beat.** A content delivery network, denial-of-service scrubbing at the edge, and outbound email deliverability. Each is a business somebody else already runs better than you will, and each is cheap next to what it replaces. Ownership is a means, not a principle.
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BEFORE YOU TOUCH ANYTHING

Every Move in this book changes a plan, a commitment or a running system. Most are reversible for a stated window, a few only until a particular step, and 1 are not reversible at all. The difference between a migration and an incident is almost never the technique - it is whether somebody worked out the way back before they started, and whether anybody had ever tested it.

READ THE ROLLBACK FIRST

Not the runbook. The Rollback section of every Move names the point of no return: the step after which the old system can no longer serve traffic or no longer holds current data. Read it, decide whether you are willing to reach that step today, and only then start at step one.

KEEP THE OLD THING RUNNING, AND KEEP PAYING FOR IT

The instinct after a successful cutover is to delete the source and book the saving. Keep the source for the window stated in that Move, with its backups. The Reversible field can also name a condition, such as an unsigned order; read it with the rollback. Once the destination accepts writes, a warm source is stale unless those writes are carried back. Recovery may need a restore and reconciliation before traffic can return.

WRITE DOWN THE STATE YOU CANNOT RECREATE

Before Stage 4, list every store whose contents exist nowhere else: the primary database, the uploads bucket, the secrets, the certificate private keys, the recovery codes for the accounts that own the domain. Everything else is rebuildable from a repository. That list is short, and it is the only part of the estate where a mistake is permanent.

TWO PEOPLE, ONE KEYBOARD

Every High-risk Move in this book assumes a second person who is not typing, is reading the runbook aloud, and has the authority to stop. It is the cheapest safety control in infrastructure and the first one teams drop when they are behind schedule. With two engineers on the whole programme, it is also the one you are most tempted to skip.

A BACKUP NOBODY RESTORED IS NOT A BACKUP

Before Stage 4, restore the backup into isolation, with outbound jobs and subscriptions disabled. Validate the expected data at its recorded recovery point and test the recovered application. A changing production database is not a valid comparison for an older backup. Record recovery time and confirm the copy and its keys survive loss of the source site.

CUT OVER WHEN YOU CAN UNDO IT, NOT WHEN YOU ARE CONFIDENT

Confidence is not evidence. The right time to move traffic is when the rollback has been rehearsed end to end, on the same day of the week, with the same people awake. A Tuesday morning with the team at their desks beats a Saturday night with one person and a laptop, every time.

VERIFY BY COUNTING, NOT BY LOOKING

A migration is finished when the destination answers the same questions as the source with the same answers. Row counts, checksums, a replayed hour of real traffic compared response by response. A page that loads is not evidence, and neither is an application that starts.

STOP WHEN YOU ARE SURPRISED

Not when something breaks - when something is merely unexpected. A row count that is off by eleven, a certificate with the wrong issuer, a replica that caught up faster than it should have. Surprise means the model in your head and the system in front of you have diverged, and continuing is guessing.

These are operational practices, not a warranty. The runbooks in this book were written against the services and versions named in each Move and will drift as those change. Rehearse every Move against a copy of your own estate before you run it against the real one, and treat any figure here as the shape of an answer rather than a quotation for your account.

WHAT IT ACTUALLY COSTS

The model keeps the existing team, salary and 160 monthly hours in every option. Cloud ops transitions to on-prem work; provider charges cover physical support. This prices a container estate at one site with off-site recovery. Add VM support, licences, backups and any required second site. Validate operating hours in Move 03. Totals exclude transition labour, overlapping cloud bills, financing and taxes.

ON AWS, COMPUTE ALONE

1,024 vCPU priced at the model's m7i Linux rate in US East, on demand, is **\$37,675** a month before a single gigabyte of storage, a load balancer or a byte of egress. Egress is the line that ends most arguments: 100 TB a month is \$7,980, every month, for the privilege of your own traffic leaving.

RENTED BY THE MONTH

\$11,700 buys the reference fleet, including power, space, networking and physical maintenance. Your existing team runs the platform with the same hours and salary, so potential staffing savings remain excluded. With retained services, the cloud bill becomes **\$20,880** a month. Renting preserves capital and avoids a facility lease. Confirm delivery, bandwidth, support and notice terms against your quote.

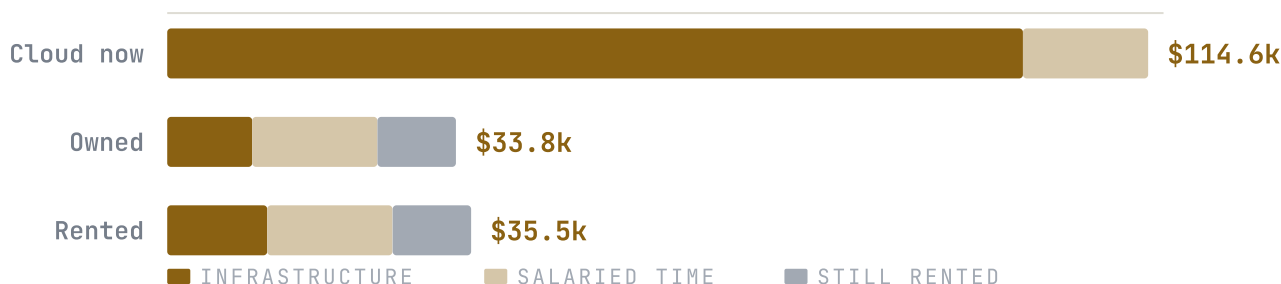
OWNED, IN COLOCATION

\$9,935 of infrastructure covers amortised hardware, spares, rack, power, transit, cross-connects and remote hands. Add \$0 of additional staffing and \$9,180 of retained services: the edge, mail, scrubbing and later Moves. Those three are **\$19,115** a month on the bill, with the existing salary on both sides. Your team takes on the platform; remote hands handles cabling and drive swaps. On-premises roles can cost less than cloud specialists, but no salary saving is assumed.

WHAT THE DIFFERENCE BUYS

\$80,885 a month, or \$970,620 a year, for a capital outlay of \$248,000 — 18 machines and a pair of switches. Capital divided by the steady-state saving is about 3 months; actual payback also includes migration and overlap. That is 71 per cent of a fully loaded \$114,644, with the salary counted on both sides; on the infrastructure line alone, which is what every other comparison quotes, it is 81 per cent. The dollar saving is the same; counting the existing salary increases the baseline.

ONE ESTATE, THREE WAYS, PER MONTH



Every AWS figure is a public list price for us-east-1 observed while writing, before any Savings Plan, private pricing agreement or credit. Hardware and facility costs are planning allowances requiring matched quotes. The comparison is a method you can repeat against your own invoice. Replace each cloud line with its effective rate; a compute discount does not also discount traffic or salary.

FIVE YEARS, THREE WAYS

Both metal options cut the cost of this reference estate over the life of the machines. Renting keeps the server purchase off the opening cash flow; colocation trades that purchase for lower monthly infrastructure costs. The table separates capital from running costs. All three options retain the same operations hours and salary; physical support is priced in provider charges and potential staffing savings are excluded. Choose the cash flow that suits the business.

THE SAME COMPUTERS, THREE WAYS, OVER 5 YEARS

	CAPITAL	RUNNING, A MONTH	TOTAL OVER 5 YEARS	AGAINST THE CLOUD
Cloud	–	\$114,644	\$6,878,613	–
Rented metal	–	\$35,524	\$2,131,413	\$4,747,200 (69%)
Colocation	\$248,000	\$29,692	\$1,992,293	\$4,886,320 (71%)

In this model, colocation costs \$139,120 less than rented metal over five years. That includes lower infrastructure costs and \$37,200 of estimated fleet value at the end, counted at fifteen per cent of the purchase price. Renting preserves that opening capital and puts physical maintenance with the provider. Replace both quotes and the residual estimate with your own; location, bandwidth and support can change which route wins. These steady-state totals exclude transition labour, overlapping cloud bills, financing and taxes.

WHAT REPLACES WHAT

Every Move names the real service on all three clouds and the one thing that differs on each, so this table is the map rather than the content. Find the row you are paying for and then read the Move. The right-hand column is the point of the whole book, and where it says keep paying, that is a conclusion rather than a gap.

AWS	GOOGLE CLOUD	AZURE	WHAT YOU RUN INSTEAD
EC2	Compute Engine	Virtual Machines	KVM virtual machines on Proxmox VE hosts
EC2 Auto Scaling	Managed instance groups	Virtual Machine Scale Sets	Fixed capacity, a priority class, and a spare on the shelf
EKS	GKE	AKS	Talos and Kubernetes, on hardware or in Proxmox VE guests
ECS on Fargate	Cloud Run	Container Apps	Deployments and Jobs on your own nodes
AMI	Machine images	Managed images	KVM guest templates; rebuild or rehearse a supported disk import
VPC	VPC networks	Virtual Network	Two switches, separate address ranges, and a documented routing plan
EBS	Persistent Disk	Managed Disks	Proxmox VM disks on Ceph RBD; Rook volumes for Kubernetes
EFS	Filestore	Azure Files	CephFS under Rook, or a filesystem the application stops needing
S3	Cloud Storage	Blob Storage	Ceph's S3 gateway - deep archive stays where it is
RDS for PostgreSQL	Cloud SQL for PostgreSQL	Database for PostgreSQL	PostgreSQL under CloudNativePG, on local NVMe
ElastiCache	Memorystore	Azure Cache for Redis	Valkey for the half that is really a cache
SQS and SNS	Pub/Sub	Service Bus	NATS JetStream, or a table in the database you already have
Glacier Deep Archive	Archive storage class	Archive tier	Keep paying where retrieval, retention and migration costs justify it
AWS Backup	Backup and DR Service	Azure Backup	Whole-VM backups, Velero and database archives, held off-site
Application Load Balancer	Cloud Load Balancing	Application Gateway	Envoy Gateway behind a virtual IP that fails over

WHAT REPLACES WHAT II

The rows that say keep paying are set in red because they are the most important lines in the table. A content delivery network, scrubbing capacity at the edge and outbound mail deliverability are not technical problems you have not solved yet. They are businesses, and you are not in them.

AWS	GOOGLE CLOUD	AZURE	WHAT YOU RUN INSTEAD
ACM	Certificate Manager	Key Vault certificates	cert-manager, with renewal failures and expiry monitored
Route 53	Cloud DNS	Azure DNS	Managed authoritative DNS; rehearse changes separately from cutover
NAT Gateway	Cloud NAT	NAT Gateway	Node addresses, and one translation hop for the few that need it
CloudFront	Cloud CDN	Azure Front Door	Keep paying - change vendor, not model
AWS Shield and WAF	Cloud Armor	Azure DDoS Protection and WAF	Keep paying for scrubbing - the capacity is upstream of you
SES	No first-party equivalent	Communication Services	Keep paying for outbound - inbound can come home
ECR	Artifact Registry	Container Registry	A pull-through cache, then your own registry of record
Secrets Manager	Secret Manager	Key Vault	Encrypted in the repository, with the key held outside the cluster
CodeBuild	Cloud Build	Azure Pipelines	Isolated build runners, or the existing hosted CI service
CloudWatch metrics	Cloud Monitoring	Azure Monitor	Prometheus, with a cardinality budget agreed in advance
CloudWatch Logs	Cloud Logging	Log Analytics	Loki, sized against retention somebody actually chose
CloudWatch alarms	Cloud Monitoring alerting	Azure Monitor alerts	OneUptime - alerting, on-call rotas, incidents and a status page
Cost Explorer	Cloud Billing reports	Cost Management	A model built from invoices, counting power, spares and the rota
Organizations	Organisation and projects	Subscriptions and Entra	Retire empty accounts; keep those that own retained services or records

WORK OUT WHETHER TO DO IT AT ALL

DONE WHEN You know what you spend, what you would spend instead, and whether it is worth it.

Whether to do this at all. Three Moves of arithmetic against your own invoice, and a fourth that names what stays rented. The stage most likely to end the programme, and that is a success rather than a failure.

☐	01	The bill, and the three lines that are most of it	0 min	24
☐	02	What you actually run	0 min	26

☐	03	The number that decides it	0 min	28
☐	04	The three things you keep renting	0 min	30

ORDER THE HARDWARE AND SIGN THE SPACE

DONE WHEN The machines are on order and the cage is signed.

What to order and where to put it. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy. Both clocks start in this stage, and they run in parallel.

☐	05	From rented vCPUs to cores you own	0 min	34	☐	07	A cage, not a data centre	0 min	38
☐	06	Sixteen machines, and the two on the shelf	0 min	36	☐	08	The order, and the weeks you cannot compress	0 min	40

TURN THE BOXES INTO A CLUSTER

DONE WHEN A cluster that could take production traffic, and never has.

Everything between a racked machine and a cluster that could take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

☐	09	Racking day	0 min	44	☐	11	The platform your workloads need	0 min	48
☐	10	The network, and the way back in when it breaks	0 min	46	☐	12	Disks: what goes local, what goes on Ceph	0 min	50

MOVE THE APP, THEN THE DATA

DONE WHEN Everything runs on your machines, and the cloud copy is still warm.

The application first, then the state. The stage that can lose a company its data, and therefore the one with the longest overlap windows, the most rehearsal and the fewest one-way steps.

☐	13	Images, secrets and one-command deploys	0 min	54
☐	14	The first service, end to end	0 min	56
☐	15	Buckets, cache and queues	0 min	58
☐	16	Postgres, the one that matters	15 min	60

CUT THE TRAFFIC OVER, AND KEEP IT ALIVE

DONE WHEN Users reach your machines, you can carry it at 03:00, and the cloud bill is zero.

The traffic, the pager and the last account. Every article about leaving the cloud stops before this stage, and every migration that regrets itself failed inside it.

☐	17	The front door	0 min	64	☐	19	Backups you have restored, and the pager	0 min	68
☐	18	Go-live, and how you abort	10 min	66	☐	20	Closing the account	0 min	70

DECIDE

The invoice, the inventory and the comparison with the salary line in it. Finish the measurements before deciding; stopping is an outcome.

BUY

The specification, the machines, the space and everything with a lead time. The stage where a mistake costs a lorry rather than a deploy.

BUILD

Racking, the network, the cluster and the disks. Nothing here moves a workload, and everything here decides how the rest goes.

MOVE

Images and secrets, the first service, the buckets and the database. The longest overlaps and the most rehearsal in the book.

RUN

The edge, the cutover, the backups you have restored, the pager, and the account you finally close.

Every Move states its cutover in minutes of user-visible downtime, its risk as blast radius rather than difficulty, and the conditions for retaining or retiring the source. 18 of the 20 Moves in this book plan for zero downtime; reference cutovers total 25 minutes, subject to each Move's prerequisites and a timed rehearsal.

WHICH MOVE DO I NEED

Nobody opens a book like this at Move 01. They open it because something on the bill, or on the pager, has become intolerable. This is the index for the way the question actually arrives.

The bill went up and nobody can say why

- 01 The bill, and the three lines that are most of it
 02 What you actually run 03 The number that decides it

Egress is now one of the largest lines on the invoice

- 01 The bill, and the three lines that are most of it
 14 The first service, end to end 18 Go-live, and how you abort

We have no idea what we are actually running

- 02 What you actually run
 13 Images, secrets and one-command deploys 20 Closing the account

The hardware sounds like a full-time job nobody has

- 07 A cage, not a data centre 09 Racking day
 11 The platform your workloads need

We want remote hands to handle the hardware calls

- 07 A cage, not a data centre
 19 Backups you have restored, and the pager
 06 Sixteen machines, and the two on the shelf
-

Somebody has asked what would happen if we just left

- 03 The number that decides it 04 The three things you keep renting
 07 A cage, not a data centre

The database costs more than the engineers who query it

- 05 From rented vCPUs to cores you own
 12 Disks: what goes local, what goes on Ceph
 16 Postgres, the one that matters

We are being asked to do this and we have two engineers

- 03 The number that decides it
 06 Sixteen machines, and the two on the shelf
 19 Backups you have restored, and the pager

Something has to move this quarter and nothing may break

- 14 The first service, end to end 17 The front door
 18 Go-live, and how you abort

We were told we cannot leave, and nobody has checked

- 04 The three things you keep renting 15 Buckets, cache and queues
 20 Closing the account
-

I DECIDE

Whether to do this at all. Three Moves of arithmetic against your own invoice, and a fourth that names what stays rented. The stage most likely to end the programme, and that is a success rather than a failure.

4	4	0	0
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

01

THE BILL, AND THE THREE LINES THAT ARE MOST OF IT

The invoice records what you were charged, not what you use. Read it to the line, find the three that are most of it ...

04

THE THREE THINGS YOU KEEP RENTING

Three capabilities do not come home: the edge, outbound mail and volumetric scrubbing. Naming them now keeps the plan ...

02

WHAT YOU ACTUALLY RUN

The estate feels infinite because nobody has written it down. Give every workload a row and an owner, then keep its ...

THE BILL, AND THE THREE LINES THAT ARE MOST OF IT · WHAT YOU ACTUALLY RUN · THE NUMBER THAT DECIDES IT · THE THREE THINGS YOU KEEP RENTING

■ ■ ■ 3 LOW RISK

■ ■ 1 MEDIUM RISK

01 THE BILL, AND THE THREE LINES THAT ARE MOST OF IT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DECIDE	—	LOW	IMMEDIATELY

The invoice records what you were charged, not what you use. Read it to the line, find the three that are most of it, then start the clock on working set.

AWS	Cost Explorer	hourly detail is a paid opt-in with a 14-day window; resource-level daily detail is a separate setting.
GOOGLE	Cloud Billing reports	a new BigQuery export in the US or EU multi-region backfills the current and previous month; regional datasets start when enabled.
AZURE	Cost Management	amortised exports support Enterprise Agreement and Microsoft Customer Agreement accounts, but management-group exports omit amortisation and commitment purchases.

BEFORE YOU START

ACCESS

- Billing export rights in every cloud account, organisation account-list access, and permission to install an agent on production instances

SOFTWARE

- The provider tools aws, gcLoud and az, with their tested versions recorded
- The chosen memory agent, with its tested version and collection configuration pinned
- A spreadsheet and a time-series store with three weeks of retention

PEOPLE

- Whoever signs the invoice, for half an hour at the end

WHY THIS WORKS

The bill and the workload measurements answer different questions. A bill records what was charged — allocation, commitment, a support percentage — not what the machines were doing. Group compute, state and traffic, keep shared charges visible, and rank the workloads inside them. Start guest memory measurements alongside the billing work: allocated RAM is not working set. A fortnight is the reference window; extend it or replay measured peaks if month-end jobs, seasonal demand or rare batches would otherwise be absent from the hardware order.

SWAP

If finance already breaks the bill down by service, start there and spend the saved time on step four. Reconcile its categories and totals before relying on them.

DO IT FASTER

Pull three invoices rather than one. One month hides a reservation renewal and a support tier that stepped up.

WATCH OUT

Metric charges depend on the collection path: custom time series, ingested bytes, samples and API reads can each be billable. Name the metrics and labels you need, then price that collection before enabling it across the fleet.

LEFTOVERS

The agents, export jobs and extra ingestion bill on until somebody removes them. Diary it for the day the hardware order is signed.

—	—	—	0 min	3 days	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing, yet. This Move may add a temporary metrics charge; calculate it from the chosen collection path and remove it when the measurements are complete.

THE RUNBOOK

- 01 Export a full month's detailed cost from every account: AWS Data Exports CUR 2.0 with resource IDs, Google Cloud's detailed BigQuery export, and `az costmanagement export` at supported subscription scopes. Record missing history; a new export may not contain last month. Keep the invoice beside it for credits, support and purchases that an export can omit.
- 02 Preserve the raw rows, then aggregate the month by resource or workload in the spreadsheet. Tag compute, state or traffic: instances and containers; databases, caches and object storage; egress, balancers and the CDN. Keep support and unallocated commitments in a shared-cost row; reconcile the grand total to the invoice.
- 03 Sort the aggregated workloads by amount and mark the largest three as the first candidates. Keep the rest in the inventory: a cheap identity service or queue can still block an expensive migration.
- 04 Reconcile the export against what you believe you run. Enumerate the accounts first — `aws organizations list-accounts`, `gcloud billing projects list` against each billing account, `az account list` — and check permissions cover the whole estate. Investigate unowned lines rather than treating them as proven waste.
- 05 Install the pinned memory agent where it is missing and collect CPU and guest memory in one time-series store. Confirm units, sampling intervals and working-set definitions before combining the three clouds' metrics.
- 06 Open the window for a fortnight, and record what resists measurement: burstable credits, page cache against anonymous memory, build minutes that were never an instance. Extend the window or test omitted peaks before Move 05 approves capacity.

ROLLBACK

The billing queries do not change workloads; an agent rollout does, so test its overhead and retain the prior configuration. Backing out removes the added agent and stops the export job. There is no point of no return here; the hardware order comes later. Preserve collected samples, mark any gaps and extend the window when needed. If the time-series store is lost, restore it.



DECIDE • 0 MIN • LOW RISK

Nothing, yet. This Move may add a temporary metrics charge; calculate it from the chosen collection path and remove it when the measurements are complete.

02 WHAT YOU ACTUALLY RUN

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DECIDE	—	LOW	IMMEDIATELY

The estate feels infinite because nobody has written it down. Give every workload a row and an owner, then keep its resources underneath it.

AWS	Resource Explorer	indexes per Region behind one aggregator and searches one account until the organisation-wide view is on, so the second account stays invisible.
GOOGLE	Cloud Asset Inventory	the caller needs <code>cloudasset.assets.searchAllResources</code> at the organisation scope; a project-level grant says nothing about the siblings on the same invoice.
AZURE	Resource Graph	reads only subscriptions the caller can already see, and pages at 1,000 rows, so an unpaginated query returns a truncated estate that looks complete.

BEFORE YOU START

ACCESS

- Read-only access to every cloud account, project and subscription, including the forgotten ones
- The billing export from Move 01, at resource granularity rather than by service

SOFTWARE

- The provider's asset CLI — `aws`, `gcloud` or `az` — and one shared document with a row per resource

PEOPLE

- The engineer who would notice each service stopping, named on every row

WHY THIS WORKS

Leaving the cloud is hard to price until somebody can say what has to move. A bill does not reveal service boundaries, dependencies or who owns a workload. Make a short service summary backed by the complete resource list; the spend alone cannot tell you how many rows there will be. Every later Move then has something finite to count against. The inventory itself changes no running system. Its risks are omission and mistaken ownership, so reconcile the provider APIs with billing, configuration and the people who operate the services.

SWAP

Where accounts are few and tags honest, the provider's tag report is a quicker first pass. Untagged resources still have to be found by hand.

DO IT FASTER

Reconcile before assigning owners. The bill sorts rows by cost, which is the order worth arguing about.

WATCH OUT

A contractor's cloud resources appear if the type and account are covered; software inside a VM generally does not. Compare the resource list with deployment configuration and guest-level services.

LEFTOVERS

A snapshot outlives the volume it came from and bills long after the rest of its account has gone. So does a retained log group.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The fourth pile, once owners, dependencies, backups and retention checks agree: orphaned volumes, unneeded snapshots, idle load balancers and proof-of-concept resources. Keep required recovery copies even when their original disks are gone.

NEEDS 01 The bill, and the three lines that are most of it

THE RUNBOOK

- 01 Pull the machine-readable list at every account's full scope: `aws resource-explorer-2 list-resources`, `gcloud asset search-all-resources` or `az graph query`. Check indexed regions and supported resource types. Follow every continuation token; Resource Explorer's separate search operation stops at 1,000 total matches. Keep one row per resource.
- 02 Check scheduled jobs, queues, DNS records and third-party webhooks against that list. Inventories include supported cloud resources, but guest-level schedulers and external services need application configuration and their owners to find them.
- 03 Reconcile against the billing export from Move 01 both ways. A bill line with no row is unowned; a row with no bill line is free, or in an account nobody has opened.
- 04 Hunt what hides: snapshots whose disk is gone, volumes attached to nothing, load balancers with no target, staging at full size overnight, the accounts opened for a proof of concept in 2022.
- 05 Label each workload VM, container or managed service. For each VM record its operating system, CPU architecture, boot mode, attached disks, licences and metadata credentials; mark which disks contain persistent state. A VM stays a VM on Proxmox VE in Move 11. Moving an application into containers is a separate project, not an unstated prerequisite.
- 06 Sort every row into one of four piles — moves as-is, moves with work, stays rented, gets deleted. Before deleting, confirm ownership, dependencies and retention needs, inspect at least thirty days of metrics and preserve a tested backup of required state. Record a destination and data-transfer method for every row that moves; check image-export restrictions before choosing that route.

ROLLBACK

Nothing here touches a running system, so backing out is discarding a document nobody acted on. The point of no return is the fourth pile: once a resource on it is deleted, getting it back means a restore from whatever snapshot survives, and for a log group there may be none. Keep the inventory in version control.



DECIDE • 0 MIN • LOW RISK

The fourth pile, once owners, dependencies, backups and retention checks agree: orphaned volumes, unneeded snapshots, idle load balancers and proof-of-concept resources. Keep required recovery copies even when their original disks are gone.

03 THE NUMBER THAT DECIDES IT

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
DECIDE	—	MEDIUM	IMMEDIATELY

Three columns, the same operations team and five years of costs. Put colocation and rented metal beside the cloud bill and choose the margin worth moving for.

AWS	AWS Pricing Calculator	two tools now carry that name, and only the one inside the billing console can see the Savings Plans your account already holds.
GOOGLE	Google Cloud Pricing Calculator	billing-account pricing requires sign-in; sustained-use discounts depend on the machine series and do not stack with committed-use discounts.
AZURE	Azure Pricing Calculator	negotiated rates require sign-in and the matching billing agreement; both Enterprise Agreement and Microsoft Customer Agreement price sheets are supported.

BEFORE YOU START

ACCESS

- Three recent invoices at the rate you actually paid, not list
- Quotes for sixteen active machines, two spares and a full rack, plus an equivalent dedicated-server rental

SOFTWARE

- One spreadsheet, three columns: cloud, rented metal and colocation

PEOPLE

- The existing cloud operations lead, to map the team's work onto the new platform

WHY THIS WORKS

Moves 01 and 02 produced a grouped bill and an inventory that accounts for it. This Move compares three ways to run that estate: the cloud, rented dedicated servers and owned hardware in colocation. Both metal options can keep the existing operations team while reducing the infrastructure bill; renting also keeps the hardware purchase off the opening budget. Put support, retained services and one-time migration work into the comparison, then apply a margin chosen before the figures. Two days of analysis gives the team a decision it can fund and defend.

SWAP

Price the three largest workloads for an initial screen, then include retained services, shared costs and labour before approving a budget.

DO IT FASTER

Use one sheet with the same capacity, support and retained-service assumptions in every column; compare it with the cloud rate you actually pay.

WATCH OUT

Equal recurring hours assumes an existing production team, a standardised platform and contracted physical support. For VMs, include hypervisor updates, guest OS patching and restore drills when validating that assumption. Vary hours, rental rates and hardware life in the sheet; setup and migration are one-time work.

LEFTOVERS

The committed-spend agreement bills to the end of its term, and that renewal date is the deadline.

—	—	—	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. This Move produces a number and, with it, permission to act or to stop.

NEEDS 01 The bill, and the three lines that are most of it 02 What you actually run

THE RUNBOOK

- 01 Write the decision rule in the spreadsheet before any figure goes in: a metal option must land at least a third under the current cloud cost to justify this programme. Apply the same margin to rented metal and colocation.
- 02 Fill the cloud column from the three invoices at the rate you actually paid, discounts in and tax out across all columns. One monthly number, checkable against Move 01.
- 03 Fill the colocation column from hardware and facility quotes: sixteen machines and two spares over sixty months, rack, power, transit, cross-connects and remote hands. Quote equivalent rented metal with traffic and hardware support. Match availability and recovery objectives: a single rack cannot replace independent cloud zones without changing that promise. Add any second site, VM licences, hypervisor support and off-site backup capacity the estate requires.
- 04 Carry the existing operations salary into all three columns. The reference assumes 160 hours a month on the same salary: cloud operations engineers take on platform work, with contracted physical tasks priced in remote hands or the rental line. Validate recurring hours, skills and on-call cover in the pilot, then budget measured differences. Charge setup, training and migration once. The published figures describe a container reference estate; price the actual VM mix separately.
- 05 Add the services that stay rented. Move 04 identifies the edge, outbound mail and scrubbing; later Moves retain archived object storage, a registry, a queue and an off-site backup copy. Their cost is already in the cloud invoice and must remain in both metal columns.
- 06 Total the three columns and apply the rule. If either metal option clears it, compare the five-year totals and the monthly cash requirement. If neither clears, keep the sheet and revisit it when the estate or quotes change.
- 07 Choose how to fund the move. Colocation buys the fleet and carries its value forward; rented metal pays monthly and keeps that capital available. Put the actual purchase quote, one-time migration effort and overlap with the cloud bill in front of whoever signs. Move 07 settles the facility and rental terms.

ROLLBACK

Nothing is ordered and nothing signed, so this Move reverses by changing a cell. The point of no return is the order in Move 06 and the contract in Move 07. Keep dated versions of the sheet, so earlier assumptions can be restored when somebody asks how the decision was reached.



04 THE THREE THINGS YOU KEEP RENTING

CUTOVER

0 MIN



LAYER DECIDE	LEAVING NOTHING — THIS MOVE DECIDES WHAT STAYS RENTED	RISK LOW	REVERSIBLE UNTIL A CONTRACT IS SIGNED
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Three capabilities do not come home: the edge, outbound mail and volumetric scrubbing. Naming them now keeps the plan honest.

AWS	CloudFront and Shield	Shield Advanced starts at \$3,000 a month on a twelve-month term; a protected distribution can front a rack, but does not protect direct rack traffic.
GOOGLE	Cloud CDN and Cloud Armor	Cloud CDN is a mode on a load balancer backend, so a colocated origin must first become an internet network endpoint group.
AZURE	Front Door and its web application firewall	Standard supports custom rules only; Premium includes managed rule sets, and the firewall policy is a separate resource.

BEFORE YOU START

ACCESS

- The three invoice lines for edge, mail and mitigation, and authority to sign

SOFTWARE

- The comparison from Move 03, open for a new permanent line
- The sending domain's current SPF, DKIM and DMARC records

PEOPLE

- Whoever owns the marketing sends, who shares the reputation
- An engineer willing to name the fourth item honestly

WHY THIS WORKS

A plan that pretends everything comes home can fail when password resets stop arriving or an attack fills the transit link. Keep three capabilities in the budget: distributed content delivery, outbound mail delivery and volumetric protection upstream of the rack. Existing providers may already serve the new origin; a hardware migration does not itself require changing them. Check coverage, capacity and terms now, and carry the quoted cost into the decision. Two days is the reference planning effort, not a promise about onboarding time.

SWAP

One vendor can cover delivery and mitigation, reducing contracts while putting both services behind the same provider.

DO IT FASTER

Ask each vendor for the contract and onboarding checklist in the first message.

WATCH OUT

Facility-included protection is often a null route with a better name. Get in writing what it does to your address.

LEFTOVERS

Keep the current edge serving the cloud origin until Move 18. Changing vendors can create overlapping bills; keeping the same edge avoids a second delivery contract.

\$5,200/mo	\$5,200/mo	0%	0 min	2 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Retire superseded providers only after their replacements work; any existing edge or mail service kept in the plan stays on the invoice.

NEEDS 02 What you actually run 03 The number that decides it

THE RUNBOOK

- 01 | Price the edge. Take delivery and egress figures from Move 01 and compare the existing vendor with alternatives at that volume. Confirm external-origin support, origin authentication and the rack's transit charges on cache misses.
- 02 | Price outbound mail on its own. Keep the current sending provider if it meets the requirements. Check SPF authorises its senders, DKIM signs correctly and the authenticated domain aligns with DMARC; marketing sends use a separate subdomain. Plan a staged migration only if the provider changes.
- 03 | Price mitigation upstream of your transit, the only place it works. Ask the facility and each transit vendor what is included, at what capacity and how it triggers: a hundred-gigabit attack against a ten-gigabit port is decided before it reaches your equipment.
- 04 | Confirm all three contracts before the hardware ships, including notice periods, minimum terms and activation dates. New dedicated sending IPs can need weeks of warm-up; agree the plan before shifting mail, and keep the working sender available during it.
- 05 | Write the quoted total into the comparison from Move 03 as a permanent line; the reference allows \$5,200 a month. Then name any additional retained service the business depends on, such as a payments integration.

ROLLBACK

Planning is reversible; the contractual point of no return is accepting a binding term, with cancellation charges governed by that agreement. If mail later moves, retain the old sender, authentication records and suppression lists until delivery is proved. Returning means restoring routing and valid authentication, then checking bounces and complaints. Domain reputation follows the domain; a new provider's IP reputation is a separate consideration.



II

BUY

What to order and where to put it. Nothing here is software, and getting it wrong is the only mistake in this book you cannot fix with a deploy. Both clocks start in this stage, and they run in parallel.

4	4	2	0
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

05

FROM RENTED VCPUS TO CORES YOU OWN

A cloud vCPU is a scheduling unit. Measure the workload before buying cores, then leave room for the guests and a ...

06

SIXTEEN MACHINES, AND THE TWO ON THE SHELF

Sixteen active nodes and two reserved replacements, bought for the shelf or included in the rental quote, because ...

FROM RENTED VCPUS TO CORES YOU OWN · SIXTEEN MACHINES, AND THE TWO ON THE SHELF · A CAGE, NOT A DATA CENTRE
· THE ORDER, AND THE WEEKS YOU CANNOT COMPRESS

■ 2 MEDIUM RISK

■ 2 HIGH RISK

05 FROM RENTED VCPUS TO CORES YOU OWN

CUTOVER

0 MIN



LAYER BUY	LEAVING THE VCPU AS A UNIT OF PURCHASE	RISK MEDIUM	REVERSIBLE UNTIL THE ORDER IS SIGNED
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A cloud vCPU is a scheduling unit. Measure the workload before buying cores, then leave room for the guests and a failed host.

AWS	EC2 instance types	a vCPU is one SMT thread on many x86 families but a whole physical core on Graviton, labelled the same.
GOOGLE	Compute Engine machine types	custom shapes have family-specific memory limits; supported families offer extended memory at a premium without adding vCPUs.
AZURE	Virtual Machine sizes	the B-series meters CPU against a credit balance, so a saturated workload reads low once its credits run out.

BEFORE YOU START

ACCESS

- The two-week measurement window from Move 01, working set sampled apart from CPU
- The instance inventory from Move 02, exported with family and size

SOFTWARE

- A sizing sheet other people can check, with the derate and headroom visible
- The vendor documentation for each instance family you run

PEOPLE

- Whoever signs the order in Move 06, present for the derate rather than shown it

WHY THIS WORKS

Move 01's samples become a part number here. On many x86 families two vCPUs share a physical core; that identifies the source topology, not the throughput of a replacement processor. Benchmark the workload on the proposed hardware. Existing VMs share a Proxmox VE host as KVM guests, each with its own operating system and resource allocation. You are buying a pool of capacity, not one physical machine per cloud instance. Memory, peak CPU demand and the capacity left after a host fails decide how many guests fit.

SWAP

With the window unfinished, use provisioned capacity as a provisional ceiling. It does not measure demand; finish the workload tests before approving the order.

DO IT FASTER

Group workloads with similar resource profiles, then test representative examples and the outliers separately.

WATCH OUT

Rightsizing advice optimises for a smaller rented instance, a different question from how much silicon to own.

LEFTOVERS

Drive IOPS depend on block size, queue depth, read/write mix and steady-state behaviour. Test the intended storage stack; a drive's peak specification is not the application's guaranteed rate.

—	—	—	0 min	3 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet, though the re-measurement in step two usually finds a service two sizes above its demand — shrink it and the bill falls this month.

NEEDS 01 The bill, and the three lines that are most of it 03 The number that decides it

THE RUNBOOK

- 01 Group the instance inventory by family and record threads per physical core from the vendor documentation. Graviton and many Arm shapes expose one vCPU per core; some x86 families do too. Do not halve those counts as though they were SMT threads.
- 02 Review burstable families separately. Azure B-series and EC2 t3 in standard mode fall to their baseline after credits run out — twenty per cent per vCPU on a t3.medium. T3 normally defaults to unlimited, but account settings and Dedicated Hosts differ. Check actual credit mode, throttling and charges, then re-measure unconstrained demand before sizing.
- 03 Enter the source topology in the sizing sheet, then benchmark a representative workload on the proposed CPU. Record peak demand and latency at that load; halving a vCPU count is not a performance test. Rebuild Arm applications for x86 before testing them on this reference hardware; an Arm disk image will not boot as an x86 KVM guest.
- 04 For VMs, budget guest RAM plus Proxmox VE, QEMU and storage reserves; do not sell the same RAM twice through ballooning. For containers, use working sets plus kubelet reserve, page cache and eviction thresholds. In both cases prove that the surviving hosts carry peak load after the largest host is unavailable, without counting an unplugged spare.
- 05 Benchmark per-core performance on the latency path as well as fleet throughput. PostgreSQL can parallelise some queries; others remain serial. Turn capacity, sustained write rate and recovery headroom into drives. Record guest vCPU allocations separately from physical cores, and hand the specification to whoever signs the order in Move 06.

ROLLBACK

A specification is undone by writing a better one: change the derate and reissue it. Nothing has been bought. The point of no return is the signature in Move 06, after which a short memory line is a second purchase with a lead time. Keep the sheet in version control so a revision can be restored.

06 SIXTEEN MACHINES, AND THE TWO ON THE SHELF

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
BUY	ELASTIC NODE CAPACITY AND CLUSTER AUTOSCALING	HIGH	UNTIL THE ORDER IS SIGNED

Sixteen active nodes and two reserved replacements, bought for the shelf or included in the rental quote, because iron does not autoscale.

AWS	EC2 Auto Scaling groups	regional EC2 vCPU quotas can stop scale-out below the group's configured maximum; request increases through Service Quotas.
GOOGLE	Managed instance groups	instance templates are immutable; changing machine shape needs a new template and a controlled update of existing instances.
AZURE	Virtual Machine Scale Sets	zonal Flexible scale sets use a fault-domain count of one for maximum spreading; it does not mean every VM shares one physical fault domain.

BEFORE YOU START

ACCESS

- The per-node sizing from Move 05 and the service inventory from Move 02
- Authority to commit the purchase or rental budget chosen in Move 03

SOFTWARE

- A quote comparison sheet with three vendors or rental providers priced line by line

PEOPLE

- A vendor or provider contact who will quote complete machines and support
- Whoever signs the purchase or rental commitment, with the chosen route agreed before ordering

WHY THIS WORKS

Fleet size comes from measured demand and the failures the platform must survive. Losing one equal-sized node removes a third of a three-node fleet or about six per cent of a sixteen-node fleet. That fraction says nothing about whether a particular database, quorum or storage replica survives. Buy enough usable capacity after control-plane reservations, storage overhead and a host failure, and place services across separate hosts. Gross core count alone does not prove that workloads fit or recover. For owned colocation, two spare machines are built, burned in and configured with the active fleet, then unplugged and shelved. Remote hands can put one into service while a failed machine goes through repair or return. The rented route reserves the same two machines as spare capacity in its eighteen-server quote; agree the provider's replacement response and how that capacity is brought into service before ordering.

SWAP

Quote a lease for the eighteen machines. Compare total payments, support, ownership at the end and the cost of returning the equipment.

DO IT FASTER

Ask for a stocked configuration that meets the measured capacity and bandwidth requirements; record the promised delivery date in the order.

WATCH OUT

Rails, power leads, 25 GbE optics and the support contract are quoted separately. A bill of materials listing only servers is short by thousands.

LEFTOVERS

A spare helps only if it stays current. Firmware, Proxmox and Talos versions drift. Update each spare on an isolated maintenance network, then return it to reserve; unplug owned spares. Enrol one in production only when replacing a failed host.

—	—	—	0 min	3 days	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing for months. The cloud instances and managed clusters carry production until Move 18 shifts the traffic; this Move commits the selected purchase or rental budget.

NEEDS 02 What you actually run 05 From rented vCPUs to cores you own

THE RUNBOOK

- 01 Put the per-node sizing from Move 05 beside the inventory from Move 02. Reserve capacity for host services and one failed physical machine before placing guests or containers. The reference fleet uses sixteen active nodes; the two spares are additional replacement capacity. Recalculate the workload budget for the chosen platform.
- 02 Write the specification on one page: dual-socket, 32 physical cores in total, 256 GB of ECC memory in a vendor-approved balanced layout, four 3.84 TB mixed-use NVMe drives plus a mirrored boot pair, redundant power supplies, two 25 GbE ports, and a controller speaking Redfish. Check memory bandwidth in the pilot; populating every channel can require more RAM. For Proxmox hosts, verify virtualisation and any required IOMMU support.
- 03 Send that page to three vendors for colocation, or three dedicated-host providers for rented metal. Record complete capacity, support and delivery terms in the quote comparison sheet. For a purchase, mark which components may be refurbished and quote new drives and power supplies; for rental, confirm support for the selected host operating system, virtualisation and remote management.
- 04 Include two identical spare nodes alongside the sixteen active ones. For owned colocation, burn them in with the fleet, then unplug and shelve them for remote hands. For rented metal, reserve two servers and document activation while the provider repairs a fault. Include all eighteen in the price; exclude spares from running capacity, quorum and storage replicas.
- 05 Confirm the route chosen in Move 03 against the hardware and facility quotes or the rental quote before making a binding commitment. For owned colocation, have the capital approver sign the hardware purchase order and record the ship date. For rented metal, order the quoted fleet under the agreed support and notice terms; the provider supplies the facility, so no hardware purchase or separate cage order is needed.

ROLLBACK

Before an order is accepted, both routes can be reversed by leaving the quote unsigned. The point of no return is the binding commitment: configured hardware may be non-returnable, while a rented fleet can carry minimum-term and notice charges. Check those terms before signing. An owned fleet can be resold at its then-current value; a rental ends under the provider's cancellation terms.

07 A CAGE, NOT A DATA CENTRE

CUTOVER

0 MIN



LAYER BUY	LEAVING THE REGION-AND-ZONE ABSTRACTION	RISK HIGH	REVERSIBLE UNTIL THE CONTRACT IS SIGNED
--------------	---	--------------	---

The facility supplies power, cooling, connectivity and remote hands. Your team operates the platform; rented metal includes the room and the machines.

AWS	Regions and Availability Zones	older accounts in some regions map zone letters differently; compare AZ IDs, which identify the same location across accounts.
GOOGLE	regions and zones	a subnet covers the whole region rather than a zone, and quota is granted per region, so capacity in one zone is not capacity in the next.
AZURE	regions, availability zones and zone-redundant tiers	zone support varies by region and VM series; moving an existing regional VM into a zone needs a supported migration path and downtime.

BEFORE YOU START

ACCESS

- The provisional fleet and facility quotes from Move 03, reconciled with Move 06 before signing
- Quotations from three facilities against one written specification

SOFTWARE

- A comparison sheet with monthly charge, cross-connect price and notice period side by side

PEOPLE

- A remote hands team with an agreed response time, plus two internal contacts authorised to direct work
- Whoever signs the term, before the price is agreed

WHY THIS WORKS

Colocation puts the building, power, cooling and physical support under a facility contract. Remote hands inspect equipment, replace labelled parts and move cables to an agreed runbook while the existing team operates the platform. The reference allows \$4,500 a month for space, transit, cross-connects and hands before usage-based power; obtain a local quote. Rented metal bundles the room, machines and contracted hardware support into the server price. One rack is one site failure domain, even with redundant power and switches. Match that limitation to the recovery objectives before choosing either route.

SWAP

Rented dedicated servers can shorten procurement and avoid a separate cage contract. Confirm delivery dates, private networking, failure-domain placement and replacement response. The reference holds staffing constant; validate the actual support and operating effort in Move 03.

DO IT FASTER

Ask for the standard agreement and the cross-connect list before price is discussed; a day of reading decides which clauses are worth arguing.

WATCH OUT

Off-site backups preserve a recovery copy; they do not keep services running during a site outage. Exercise the restore and price a second failure domain when the required recovery time demands it.

LEFTOVERS

The facility invoices from the agreed commencement date while cloud charges continue. Budget overlap from the delivery and migration schedule, including delays.

—	\$4,500/mo	—	0 min	4 days	3 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing. This Move adds an outgoing instead of removing one; the cloud bill does not move until the first service does.

NEEDS 03 The number that decides it

THE RUNBOOK

- 01 Write one specification — full rack, committed kilowatts, two feeds, cross-connects, remote hands, term — and send it to three facilities. Set the location around latency and physical response time. If a site outage must recover faster than replacement hardware and an off-site restore, include an independent recovery site in Move 03's budget.
- 02 Size power against measured peak draw, including switches and recovery load. The reference's sixteen nodes at 450 W total 7.2 kW before networking, with a provisional 10 kW commitment. Verify the actual fleet, continuous-load limits and cooling allowance; either feed must carry the whole rack after the other fails. Keep owned spares unplugged.
- 03 Contract the remote hands scope: inspections, labelled disk and machine replacements, cabling and power actions under an approved runbook. Put ticket-to-touch, included hours, out-of-hours rates and a named escalation in writing. Give the team the rack map and spare-part locations; name two internal contacts who can authorise work remotely.
- 04 Ask which carriers have live fibre in the meet-me room and which of them will sell at your volume, then get the cross-connect price list in writing, installation and disconnection included. Three carriers and a cheap cross-connect beat better marketing.
- 05 Negotiate the exit with the entrance: a term no longer than the hardware life, a capped escalator, a notice period you could serve. Three years with twelve months' notice repeats the mistake you are here to correct.
- 06 Before anyone signs, price the alternative in the next column: the same eighteen machines rented by the month, with power, networking and hardware replacement terms specified. Compare current quotes at matching capacity and support. Rented metal removes the fleet purchase and the separate facility agreement; use Move 03's full-cost comparison to choose the route.

ROLLBACK

Refusing a facility before signature leaves no contract to unwind. The contractual point of no return is accepting a binding agreement, which can create liabilities before the commencement date. Check cancellation, minimum-term and notice charges before signing. Keep the losing quotations and the agreed service specification so earlier assumptions can be restored during renewal discussions.

08 THE ORDER, AND THE WEEKS YOU CANNOT COMPRESS

CUTOVER

0 MIN



LAYER BUY	LEAVING PROVIDER-ASSIGNED ADDRESSES AND MANAGED TRANSIT	RISK MEDIUM	REVERSIBLE UNTIL THE ORDER IS SIGNED
--------------	--	----------------	--

Everything with a lead time goes on one order this afternoon, so the queues run beside each other instead of one after another.

AWS	Elastic IP and Direct Connect	the cross-connect authorisation expires ninety days after issue, and an Elastic IP bills on after its instance is gone.
GOOGLE	external IP and Cloud Interconnect	Partner Interconnect uses a provider's connection to Google; order access to that provider separately, with capacity from the supported list.
AZURE	Public IP and ExpressRoute	the circuit bills from creation, before the carrier has provisioned anything against its service key.

BEFORE YOU START

ACCESS

- The signed facility agreement from Move 07 and its cross-connect price list
- Purchase authority to sign transit and circuits

SOFTWARE

- The switch part numbers from Move 06, for the optic compatibility list
- A written order sheet, one promised date and one name per line

PEOPLE

- A named technical contact the carriers can reach in week three

WHY THIS WORKS

Orders have lead times that engineering effort cannot shorten. Put cross-connects, optics and carrier access into parallel queues while the machines are being built. The four-week wait below is a planning allowance; replace it with the longest confirmed delivery date. Pay for two transit paths when they can carry your addresses, and record any initial dependence on facility transit. Cancellation rights come from each quote and signed term, not from whether anything has been installed.

SWAP

Buy every line from the facility on one invoice. It costs more per megabit and takes four vendors off the critical path.

DO IT FASTER

Ask for the authorisation templates before the agreement is countersigned; no queue starts until they are lodged.

WATCH OUT

The AWS Direct Connect authorisation expires after ninety days; download a renewed letter if installation slips. Other providers have their own terms.

LEFTOVERS

Keep blended transit while it is the only route to facility-assigned addresses. Retire it only after replacement addressing and redundant transit pass their own soak.

—	—	—	0 min	3 days	4 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Blended transit can go only after its addresses and routing have been replaced, both remaining paths pass failover, and a full month's traffic has stayed healthy.

NEEDS 07 A cage, not a data centre

THE RUNBOOK

- 01 Order the cross-connects the day the facility agreement is countersigned: one per upstream, one for the out-of-band line, one spare. Ask whether the panels to the meet-me room are in place; that decides days against weeks.
- 02 Take the facility's blended transit for go-live, already in the building, and order a second upstream from a carrier on a different fibre route to run beside it. On the facility's own addresses that second circuit buys a price and a path to grow into rather than a failover, because only the facility's transit routes to you; it becomes a failover the day you announce your own prefix to both.
- 03 Take the required IPv4 space and IPv6 from the facility. If independent upstream failover is required at go-live, arrange a portable prefix, routing authorisation and an autonomous system number now; a second site is not required. Confirm both carriers accept the prefix before ordering. Facility-assigned space may require renumbering later.
- 04 Order the out-of-band line: a small separate circuit, or a mobile router. It is the cheapest item here and the one that matters at three in the morning.
- 05 Order optics and cables from the switch vendor's supported list, twenty per cent over the port count. Then chase the order sheet weekly.

ROLLBACK

Until a signature this Move is undone by an email. The point of no return is the first signed term, not the first delivery: transit and cross-connects carry minimum terms in months, and a circuit against the wrong cabinet is paid for whether or not it is lit. Keep the quotes in version control, so a position can be restored at renewal.

III

BUILD

Everything between a racked machine and a cluster that could take production traffic. Nothing here moves a workload; all of it decides how well the rest of the book goes.

4	4	3	0
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

09

RACKING DAY

Prove remote console access to the whole batch, then shelve the spares. Rails, power and labels are the work ...

12

DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH

Choose who owns each physical disk.
Proxmox supplies VM datastores; Rook supplies storage on the direct Talos path. ...

10

THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS

Address ranges written down before anything is configured, two switches either of which may fail, and a way in that ...

RACKING DAY · THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS · THE PLATFORM YOUR WORKLOADS NEED · DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH

■ 1 MEDIUM RISK

■ 3 HIGH RISK



LAYER BUILD	LEAVING THE PROVIDER'S SERIAL CONSOLE AND BOOT DIAGNOSTICS	RISK MEDIUM	REVERSIBLE IMMEDIATELY
----------------	---	----------------	---------------------------

Prove remote console access to the whole batch, then shelve the spares. Rails, power and labels are the work; recovery from outside the building is the proof.

AWS	EC2 Serial Console	requires account access and a supported instance type; it exposes a serial port, not a controller that mounts your recovery ISO.
GOOGLE	Compute Engine serial console	switched on by the serial-port-enable metadata key, and an organisation policy can forbid it estate-wide.
AZURE	Boot diagnostics and Serial Console	portal-created VMs default to managed boot diagnostics; interactive console access needs its own permissions and can be disabled subscription-wide.

BEFORE YOU START

ACCESS

- A booked day on site for two named people
- An out-of-band line that is live and independent of the uplinks

SOFTWARE

- `ipmitool` and a Redfish client on a laptop outside the building
- A memory tester, a drive exerciser and one firmware bundle for the whole batch

PEOPLE

- Two people for the lift; a chassis at head height is not a one-person job

WHY THIS WORKS

The physical work is ordered so nothing has to be undone: rails across the active positions before a chassis is lifted, weight low, power split so one distribution unit cannot take the estate. None of that is the point. The Move exists for the baseboard management controller in each node — a small computer with its own processor, port and firmware, and the whole of the console the cloud gave you. Proved from outside the building, it turns two years of faults into an afternoon at a keyboard rather than a drive to the facility.

SWAP

Quote remote hands for racking and cabling to a drawing. Keep controller access and acceptance tests with your platform owner.

DO IT FASTER

Address and password the controllers on a bench the week before, so the day itself is cabling and testing.

WATCH OUT

Controllers can share a host NIC when configured that way. Prove guest and internet traffic cannot reach them, even during a host reboot.

LEFTOVERS

Cardboard, the spare's rail kit, and the temporary address scope that hands out addresses for months.

—	—	—	0 min	6 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The active fleet draws power in one building while the whole cloud bill runs in another.

NEEDS 06 Sixteen machines, and the two on the shelf 08 The order, and the weeks you cannot compress

THE RUNBOOK

- 01 Fit the active fleet's rail pairs before any chassis goes in, checked against the cabinet's depth. Rack bottom-up, heaviest low, two people on each lift. Test the spares temporarily; their permanent home is the shelf.
- 02 A-side power from one distribution unit, B-side from the other, so a node's supplies never share a strip. Verify each feed can carry the full rack load, then pull one feed per node and confirm it stays up.
- 03 Cable data in one colour and management in another, both ends labelled before anything is plugged in. Label the position, not the machine.
- 04 Give each controller a static address on the isolated management network, serial-over-LAN on, and a unique password stored outside the cluster. Disable unused shared-NIC access. Check power status through `ipmitool` and Redfish, recording which functions and licences each controller provides.
- 05 Leave the building. From a laptop elsewhere, over the out-of-band line, watch a node POST, enter firmware setup, mount an image over virtual media and power-cycle it. Until that works from outside, nothing here is built.
- 06 Run the memory tester and the drive exerciser overnight on all eighteen nodes, level them all to one firmware bundle, and before leaving site take front and rear photographs, the port map and a file naming each active rack position and each spare. Unplug and shelve the spares after acceptance; they remain outside every production cluster.

ROLLBACK

Nothing here carries data: a mistake is undone by unranking a machine the same afternoon. The point of no return is the day this cabinet takes production traffic, after which moving a cable is a change window. Firmware is the exception — a downgrade is sometimes refused, so record the shipped version per serial and keep the vendor's image so a working level can be restored.

10 THE NETWORK, AND THE WAY BACK IN WHEN IT BREAKS

CUTOVER

0 MIN



LAYER BUILD	LEAVING CLOUD-MANAGED PRIVATE NETWORKING	RISK HIGH	REVERSIBLE IMMEDIATELY
----------------	---	--------------	---------------------------

Address ranges written down before anything is configured, two switches either of which may fail, and a way in that depends on neither.

AWS	VPC and its subnets	a subnet occupies one Availability Zone; supported EC2 paths allow a 9001-byte MTU, but internet-gateway traffic is limited to 1500.
GOOGLE	VPC networks and subnets	changing network MTU requires stopping all attached VMs, then starting them; a guest reboot is insufficient, and Windows needs an explicit interface update.
AZURE	Virtual Network and its subnets	no broadcast or multicast on the wire, so gratuitous ARP and VRRP never worked and a floating address had to be a load balancer.

BEFORE YOU START

ACCESS

- Console access to both switches, and the out-of-band line from Move 09 answering
- The public address range from Move 08, and the private ranges the organisation already uses

SOFTWARE

- `ipmitool` 1.8.19 or newer, to prove the baseboard controllers answer on the management range
- A packet test that sets the do-not-fragment bit, because a default-size ping proves nothing here

PEOPLE

- Two engineers for the switch-pull drill, one at the rack and one watching the graphs

WHY THIS WORKS

The address plan is cheap to get right today and expensive to change in a year, because by then it sits in a partner's firewall and on every machine. It is written first, on paper: separate ranges, each with twice the room the estate needs now. The switches come as a pair, since one switch is one maintenance window that takes everything down with it. MTU gets a step of its own, because a mismatch does not fail cleanly — it passes small requests and stalls large ones. Nothing here carries production traffic yet, which is what makes the week safe.

SWAP

Where the switches will not do multi-chassis aggregation, give each node an address per uplink and let it route to both independently: more host configuration, fewer shared failure modes.

DO IT FASTER

Generate both switch configurations from the address plan rather than typing them twice. Hand-built pairs differ in one port, and the drill is what finds it.

WATCH OUT

The way back in must not depend on the fabric it exists to rescue. A jump host on the data VLAN, or one hosted only inside the Proxmox cluster it must rescue, cannot recover that cluster.

LEFTOVERS

The cloud private network and its per-gigabyte charges keep running throughout, and nothing here removes them.

—	—	—	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The cloud's private network carries every service you run until the traffic moves in Stage 5.

NEEDS 09 Racking day

THE RUNBOOK

- 01 Write the address plan before touching a switch. Reserve management, storage, host, guest, Kubernetes pod and service networks, and the public block from Move 08, none overlapping the office, cloud accounts or partner routes. For Proxmox, reserve cluster communication and migration paths; storage recovery must not starve quorum traffic. Store the plan in source control.
- 02 Cable the two top-of-rack switches as a multi-chassis pair with a peer link, and run two uplinks from every node into different switches, bonded. One switch then fails as a component rather than as an outage.
- 03 Build the controller VLAN with no route to the internet. Reach baseboard controllers and Proxmox administration through the independent jump host, with access rules separating controllers from host management. Guest networks cannot reach either; host updates use controlled outbound access. Prove it from outside the building with an `ipmitool` power status on every node.
- 04 Record the IP MTU for each path. Keep bonds, bridges and guest interfaces consistent, allow Ethernet and VLAN overhead in switch frame limits, and reduce tunnel-interface MTU for encapsulation. Internet and private storage paths need not share one MTU.
- 05 Test full-size packets that refuse fragmentation on every traffic path, then application requests through tunnels. Permit the ICMP messages used by path MTU discovery. Small pings can pass while larger requests stall; there is no fixed percentage of traffic a mismatch breaks.
- 06 Run the drill in daylight, with both engineers watching. Pull the power from one switch, confirm every bond keeps a member, then plug it back in and confirm the link rejoins. Nobody who declines this today will do it under load.

ROLLBACK

Nothing here carries production traffic, so backing out is reverting one switch configuration and restoring the previous one from source control. The point of no return is a date rather than a command: once node addresses and partner firewall entries reference the plan from step one, renumbering becomes a weekend rather than an edit. Argue the ranges into shape now, not after Move 11.

11 THE PLATFORM YOUR WORKLOADS NEED

CUTOVER

0 MIN



LAYER BUILD	LEAVING MANAGED VM AND KUBERNETES COMPUTE	RISK HIGH	REVERSIBLE IMMEDIATELY
----------------	--	--------------	---------------------------

Cloud VMs become KVM guests on Proxmox VE. Containers keep Kubernetes. Owning the hardware does not require changing the application's packaging.

AWS	EC2 and EKS	instance-role credentials depend on metadata, while pod identity can depend on an IAM OIDC trust that your new cluster must replace.
GOOGLE	Compute Engine and GKE	attached service-account credentials come from a metadata service; copying the boot disk does not bring that service with it.
AZURE	Virtual Machines and AKS	managed identities are Azure resources with role assignments; neither a guest image nor a pod manifest transfers that trust.

BEFORE YOU START

ACCESS

- The machines and VLANs from Move 10, with host roles assigned from Move 02's inventory
- An API virtual address for Kubernetes and management access restricted to the private VLAN

SOFTWARE

- For VMs: the Proxmox VE 9.2-1 installer, its published checksum, and versioned guest OS templates with a matching QEMU guest agent
- For Kubernetes: Talos 1.14.0 and Kubernetes 1.35.8 pinned in Git; matching talosctl and kubectrl clients
- helm, Cilium chart 1.20.1, the cilium CLI, and cert-manager chart v1.21.1
- An encrypted credential store outside the platform for Talos configuration and recovery keys

PEOPLE

- A platform owner and someone to disconnect a host while the owner watches recovery

WHY THIS WORKS

Bare metal is the hardware layer. For an estate with VMs, install Proxmox VE on the physical hosts and run those workloads in KVM guests. A guest keeps its operating system, service manager and deployment tooling; several guests share each host. Proxmox manages their lifecycle and placement. It does not recreate the cloud's metadata, identity or autoscaling services. For an estate already running containers, Talos Linux directly on the hosts remains a useful choice. In a mixed estate, Talos can also run in Proxmox guests, with physical failure domains and storage ownership made explicit. The strip below retains the container reference's control-plane charge. Add VM platform support, guest licences and operating time to Move 03 before using that model for a VM estate.

SWAP

A container-only estate can skip Proxmox and boot Talos directly. Choose from the inventory, not from the name of the cloud service on the invoice.

DO IT FASTER

Build a guest template once and clone it. Keep its image digest and configuration alongside the application release.

WATCH OUT

Do not stack Rook OSDs on replicated virtual disks. Passing through physical storage restricts migration; Move 12 assigns each device one owner.

LEFTOVERS

Schedule host and guest patching separately, and rehearse upgrades on the spare. A VM snapshot is not an independent backup.

\$220/mo	\$0/mo	100%	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. Cloud instances and managed clusters still serve production. Their charges stop only after their workloads pass the later migration gates.

NEEDS 10 The network, and the way back in when it breaks

- 01 Write the host and guest plan in Git. Reserve CPU and RAM for the hypervisor, storage and a failed host as sized in Move 05. Enable hardware virtualisation in firmware for the VM pool. Keep the management and cluster-quorum networks away from guest traffic and storage saturation.
- 02 For that pool, verify the Proxmox VE 9.2-1 checksum and each empty boot pair's serial before the installer erases it. Apply security updates in the lab, record the tested package versions, then roll that set to the hosts. Create a Proxmox cluster and join the empty hosts; use at least three for quorum.
- 03 Create test KVM guests from the versioned OS templates on the local datastore reserved during host installation. Set CPU, RAM, boot mode, VirtIO disks and VLAN bridges; enable and test the QEMU guest agent. Pick a CPU model supported by every migration destination. Keep HA disabled: Move 12 moves ordinary guest disks to shared storage before production.
- 04 Where Kubernetes is needed, allocate three control-plane nodes on separate physical hosts, whether they are Talos guests or physical machines. For Talos guests, include the `siderolabs/qemu-guest-agent` extension in the pinned installer image. Distribute workers across hosts and reserve etcd memory and disk I/O. Guests on one host share its failure; keep capacity available for recovery.
- 05 Generate configuration with `talosctl gen config --kubernetes-version 1.35.8`. Keep generated files and their private keys in the encrypted store; commit only non-secret patches. Confirm each installation disk before Talos repartitions it. Set the API virtual address, disable the default CNI and kube-proxy, apply configuration, then run `talosctl bootstrap` on one control plane.
- 06 Install Cilium with `helm install` at chart version 1.20.1, using its documented Talos capabilities and cgroup settings. Set `ipam.mode=kubernetes`, `kubeProxyReplacement=true`, and the API to KubePrism at `localhost:7445`. Verify kube-proxy is absent with `kubectl`, then run the cilium connectivity test. Save the complete Helm values in Git.
- 07 Install cert-manager with `helm install` at chart version v1.21.1, enabling `crds.enabled=true`. With `kubectl`, wait for its controller, webhook and cainjector deployments to become available. Later database and edge Moves need these certificate services. A VM-only estate skips the Kubernetes steps.
- 08 Disconnect one physical host while only test workloads run. Kubernetes must retain API quorum and reschedule replaceable pods; local guest disks remain unavailable, so expect no VM failover yet. Move 12 repeats this with shared storage and HA affinity rules. A failed VM restarts; it does not live-migrate from a dead host.

ROLLBACK

While only test guests and an empty cluster exist, restore the recorded configuration or rebuild from the saved images and encrypted keys. Returning to kube-proxy also requires a compatible Cilium configuration and another connectivity test; one flag cannot repair every network failure. The point of no return is the first production disk or application write: after that, rebuilding a host requires preserving its datastore and restoring affected workloads. Keep cloud production authoritative until the later Moves test migration and recovery.

12 DISKS: WHAT GOES LOCAL, WHAT GOES ON CEPH

CUTOVER

0 MIN



LAYER BUILD	LEAVING MANAGED BLOCK AND SHARED-FILE STORAGE	RISK HIGH	REVERSIBLE IMMEDIATELY
----------------	---	--------------	---------------------------

Choose who owns each physical disk. Proxmox supplies VM datastores; Rook supplies storage on the direct Talos path. Replication belongs in one layer.

AWS	EBS and EFS	a gp3 volume lives in one Availability Zone and its provisioned throughput is separate from capacity; EFS in bursting mode spends credits that run out.
GOOGLE	Persistent Disk and Filestore	Persistent Disk performance depends on disk type, size and VM limits; Filestore basic tiers can grow but cannot shrink, unlike zonal, regional and enterprise tiers.
AZURE	Managed Disks and Azure Files	Premium SSD v1 uses performance tiers; Premium SSD v2 configures IOPS and throughput separately, and SSD file shares use a FileStorage account.

BEFORE YOU START

ACCESS

- The device inventory from Move 09, with data drives identified by serial and boot drives excluded
- Proxmox storage administration or the Talos configuration from Move 11, according to the chosen platform

SOFTWARE

- The Proxmox-supported Ceph package version pinned across hosts, including current security fixes
- `helm` and `kubectl` for Talos or external Ceph clients, with Rook 1.20.7 and Ceph 20.2.4 pinned in the deployment configuration
- `ceph` for health, capacity and recovery output during the drive-pull test

PEOPLE

- One named owner for Ceph, including its object gateway and the recovery drill

WHY THIS WORKS

A VM needs a datastore before it needs Kubernetes. On the Proxmox path, the hosts own Ceph and present shared block storage to guests. On the direct Talos path, Rook manages Ceph on physical disks. Running another replicated Ceph inside guests whose disks already sit on host Ceph multiplies copies and hides the physical failure domains. Choose one owner per device. PostgreSQL can use local NVMe because it replicates to its own standbys. That saves a second replication layer but binds each copy to its host. Its standbys must occupy different physical machines, even when the database runs inside Kubernetes VMs. The reference disk split and savings below describe the direct Talos build; price the VM datastore and its recovery capacity separately.

SWAP

Proxmox can use existing shared storage or local datastores. Local disks need transfer during migration; host failure needs replication or restore, so demonstrate the promised recovery before choosing them.

DO IT FASTER

Name storage classes and datastores for the recovery promise they make, so each workload gets a deliberate placement.

WATCH OUT

Snapshotting a volume is not an independent backup. Keep a tested copy beyond the storage cluster that serves it.

LEFTOVERS

The managed volumes, file shares and snapshots keep billing until their contents have moved and the retention window ends.

\$9,400/mo	\$850/mo	91%	0 min	6 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing this week. Retire each managed disk or share only after its workload, data and rollback copy have moved. Object migration in Move 15 does not retire block volumes or file shares; VM disks and shared files need their own acceptance checks before Move 20 removes the leftovers.

NEEDS 11 The platform your workloads need

- 01 Record each disk's owner and confirm the selected devices are empty before provisioning. On direct Talos, reserve three data drives per participating node for Rook and the fourth for local PostgreSQL; exclude shelf spares. On Proxmox, assign host datastore disks separately from any local database devices.
- 02 For Proxmox, use its Ceph wizard with the pinned packages, monitors on separate physical hosts, and only the recorded datastore devices. Add an RBD pool as VM storage. On Talos, add the official Rook chart repository and use `helm install rook-ceph rook-release/rook-ceph --version 1.20.7`. Apply a CephCluster configuration pinning Ceph 20.2.4 and the inventoried raw disks.
- 03 Set replicated pools to three copies and `min_size 2`, with the failure domain at the physical host. Use `ceph df` and model redistribution after losing the largest storage host; keep utilisation below 70 per cent and verify recovery still has room. Guest count is not a count of independent replicas.
- 04 Provision the shared filesystem and object gateway required by Moves 15 and 16. Rook creates these on direct Talos. For host-owned Ceph, its owner provisions CephFS and a separate RADOS gateway service; an RBD datastore alone supplies neither an S3 endpoint nor its credentials. Connect Kubernetes through Rook's external-cluster mode, with restricted client keys and the existing gateway endpoint, never guest OSD disks.
- 05 Keep each local PostgreSQL copy on a different physical host, with placement rules enforced during failover. If passing a disk or controller into a guest, give it exclusive ownership and exclude it from host Ceph. That guest loses ordinary live migration; database replication and a tested restore provide its recovery path.
- 06 On Proxmox, move ordinary guest disks from Move 11's local bootstrap datastore to shared RBD. Verify every required disk and network exists on eligible hosts, then register those guests as HA resources with negative resource-affinity rules separating control planes. Exclude guests tied to local or passthrough devices. Write test data to a VM disk or a claim created with `kubectl` on Talos. Power off one physical host and verify recovery and contents, including the filesystem and gateway. Confirm no failure removes every database replica.
- 07 With only verified test data present, pull a storage drive and time recovery with `ceph -s`. Reseat it, wait for health to recover, and record the duration before accepting production data.

ROLLBACK

Before test data lands, backing out means removing the empty pools and restoring the recorded disk layout. The point of no return is the first production write: leaving either Ceph or a local datastore then requires a data migration with a write stop. Back up a test VM or volume outside this cluster, restore it elsewhere, and verify its contents before acceptance. A local database copy needs its own restored backup as well as surviving replicas.

IV MOVE

The application first, then the state. The stage that can lose a company its data, and therefore the one with the longest overlap windows, the most rehearsal and the fewest one-way steps.

4 MOVES	3 ZERO DOWNTIME	2 HIGH RISK	15 MINUTES, TOTAL
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IF YOU ONLY DO THREE

13

IMAGES, SECRETS AND ONE-COMMAND DEPLOYS

Registry, secrets and deployments are proved together in staging, before customers depend on the new platform.

15

BUCKETS, CACHE AND QUEUES

Copying bytes is the easy part. A live move also needs an ordered record of changes, retryable publishing and ...

14

THE FIRST SERVICE, END TO END

The first service runs in a VM or a container on your hardware, with its data still in the cloud and a traffic weight ...

IMAGES, SECRETS AND ONE-COMMAND DEPLOYS · THE FIRST SERVICE, END TO END · BUCKETS, CACHE AND QUEUES · POSTGRES, THE ONE THAT MATTERS

■ 2 MEDIUM RISK

■ 2 HIGH RISK

13 IMAGES, SECRETS AND ONE-COMMAND DEPLOYS

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
MOVE	MANAGED REGISTRIES, SECRET STORES AND HOSTED CI RUNNERS	MEDIUM	IMMEDIATELY

Registry, secrets and deployments are proved together in staging, before customers depend on the new platform.

AWS	ECR and Secrets Manager	registry tokens last twelve hours; scheduled secret deletion has a seven-to-thirty-day recovery window, with thirty days the default.
GOOGLE	Artifact Registry and Secret Manager	values are immutable, so every rotation leaves a billable version until destroyed.
AZURE	Container Registry and Key Vault	soft-delete cannot be switched off once enabled; purge protection holds deleted names for the configured seven-to-ninety-day retention period.

BEFORE YOU START

ACCESS

- A deployment repository, with its deploy key held outside the platform
- Registry or VM artifact access and CI credentials

SOFTWARE

- For containers: he`lm, Argo CD and Harbor at the pinned versions below
- For GitHub Actions: Actions Runner Controller and isolated build infrastructure
- For VMs: versioned deployment and runner tooling
- sops and age, with the private key held externally

PEOPLE

- One engineer responsible for the deployment and recovery conventions

WHY THIS WORKS

Containers use Harbor and Argo CD; VMs keep versioned templates and existing deployment tools. Either route must recover credentials using a key held outside the platform. VM-only estates can retain their artifact store and runners. Staging exercises production's templates, images, secrets and deployment path against its own data. A fortnight of branch deployments, plus a rebuild and restore, tests recovery before Move 14 introduces customer traffic.

SWAP

A plain registry over the same object store drops the database and cache Harbor needs, and with them projects, quotas and scanning.

DO IT FASTER

Prove the image pull path first, then switch pushes after the restore test passes.

WATCH OUT

Keep bootstrap images independent of Harbor and prove recovery from a full rack restart with test workloads. Staging must use its own data copies before step seven; a registry cache does not break every dependency loop.

LEFTOVERS

The cloud registry bills storage and egress until its images go, a Move 20 decision, and hosted runner minutes continue for anything still routed at them.

\$2,600/mo	\$260/mo	90%	0 min	7 days	2 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The hosted runner minutes, once a week of builds has passed on your own nodes with no fallback. The non-production estate's cloud compute, once the fortnight in step seven is done and nobody has asked to go back. The cloud registry and its secret store stay funded until Move 20.

NEEDS 11 The platform your workloads need 12 Disks: what goes local, what goes on Ceph

THE RUNBOOK

- 01 For containers, use `helm install` to deploy Harbor 2.15.2 at chart version 1.19.2, storing image blobs on Move 12's Ceph object gateway. Back its database separately. Configure a proxy project only where the upstream adapter supports it; test private-image authentication before repointing pulls. VM-only estates can retain their artifact store.
- 02 Create a separate normal Harbor project for pushes; a proxy-cache project cannot accept them. Move the push target, mirror retained releases and verify image digests. Keep the cloud registry current during the trial so either deployment path can use the same release.
- 03 For Kubernetes, `helm install` Argo CD 3.5.2 at chart version 10.8.4 and configure applications with automated sync and self-healing, including namespace creation. For VMs, record template checksums, resource settings and the pinned deployment recipe in Git. Bootstrap credentials stay outside the platform.
- 04 Encrypt every secret into the repository with `sops`, under an age key kept in a password manager and one offline copy. Feed decrypted secrets through the existing VM deployment tool or the Kubernetes bootstrap path. Limit the bootstrap key to that process; never bake it into a template or image.
- 05 After backing up staging data, rebuild a disposable VM from its template and deployment recipe, or delete a non-production namespace and let the reconciler recreate it. Check that the workload and its secrets return; restore the staging data and verify its contents.
- 06 For GitHub Actions, install both Actions Runner Controller and its runner scale-set chart at version 0.14.2 on isolated build infrastructure. Set the repository URL, credential-secret reference and runner image digest; route jobs by scale-set name. Other CI services and VM deployments can retain existing runners. Test a commit through build and deployment.
- 07 Move staging and review environments to their assigned guests or namespaces, update DNS, and use them for a fortnight of branch deployments. Their data may remain in the cloud, but only in staging's own stores. Reboot a guest or redeploy a container for each route in use; neither passes until its configuration survives.

ROLLBACK

Revert the deployment recipe and route builds back to the existing runners. Restore staging data before returning a workload that has accepted writes; a Git revert cannot recover those writes. The point of no return is removing the old artifacts and credentials before their replacements have passed a rebuild. Keep the cloud registry, VM artifacts and secret store available until the corresponding guest or cluster has recovered from its recorded configuration.

14 THE FIRST SERVICE, END TO END

CUTOVER

0 MIN



LAYER ➡ MOVE	LEAVING CLOUD VM AND CONTAINER COMPUTE	RISK ■ MEDIUM	REVERSIBLE IMMEDIATELY
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The first service runs in a VM or a container on your hardware, with its data still in the cloud and a traffic weight as the way back.

AWS	EC2 and ECS on Fargate	VM Import/Export excludes Marketplace images and images with encrypted EBS snapshots; its Windows and SQL Server image restrictions also block many exports.
GOOGLE	Compute Engine and Cloud Run	custom images export through Cloud Storage, but Google-provided Windows Server licences do not permit running those images outside Google Cloud.
AZURE	Virtual Machines and Container Apps	direct VHD download cannot use a disk attached to a running VM; a snapshot can be exported instead, but live snapshots are only crash-consistent.

BEFORE YOU START

ACCESS

- Proxmox management access and the VM configuration repository, or write access to the Argo CD application repository
- Permission to edit weighted DNS records, a working TLS service endpoint through the existing edge, and the required cloud credentials

SOFTWARE

- For VMs: the versioned OS template and existing application deployment tooling; qm for an optional disk import
- For containers: kubectl and argocd, pointed at the new cluster
- dig for checking public DNS during the ramp

PEOPLE

- The service owner, with test requests, an abort threshold and any guest licence approval
- Whoever maintains partner egress allowlists, warned in advance

WHY THIS WORKS

The first service produces evidence before it produces savings. Its database remains in the cloud, so the new variables are the host platform, network path and identity. For a VM workload, create a fresh KVM guest on Proxmox VE and deploy the same application using its existing tooling. No container conversion is required. For a container workload, deploy the existing image to Kubernetes. Both routes can run beside the cloud copy and accept the same traffic ramp. This Move covers stateless services, including those installed inside VMs. A writable application disk makes a VM stateful even when nobody calls it a database. Keep that data authoritative in the cloud, or defer that VM until its owner has rehearsed the data transfer, write stop and return path for Move 18. The zero-minute strip is for parallel stateless deployment, not a promise about exporting a running machine.

SWAP

A proxy you already control can shift request weight faster than DNS. Keep the same acceptance and abort thresholds.

DO IT FASTER

Reuse the verified guest template and deployment recipe for the next VM with the same operating system.

WATCH OUT

Image export preserves bytes, not cloud identity, instance-store data or attached services. Verify every dependency from the inventory.

LEFTOVERS

Exported images, temporary buckets, snapshots, SAS links and both allowlist entries need dated cleanup after verification.

\$30,980/mo	\$0/mo	100%	0 min	20 days	8 weeks
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Each old VM or container deployment after its own completed soak and return-path check. Shared instance groups, clusters and log retention stay until their last dependent service has passed; retain the snapshots required by the rollback plan.

NEEDS 13 Images, secrets and one-command deploys

THE RUNBOOK

- 01 Choose a low-impact stateless service from Move 02. Identify every disk write, scheduled job and metadata dependency. Disable duplicate jobs in the test copy, and keep it isolated from production traffic until the owner confirms it cannot create a second writer.
- 02 For a VM, create a KVM guest from the OS template and deploy the pinned application release with its existing tooling. For containers, commit the manifests and let `argocd` sync them. Keep the cloud database endpoint. Replace metadata credentials with an explicit workload identity or scoped credential, and update partner allowlists.
- 03 If rebuilding a guest is impractical, check the provider's export restrictions and licence portability first. Take an application-consistent copy of every required disk, export through the provider console, then use `qm disk import` to import a supported image. Attach the disks, match BIOS or UEFI and set boot order. Rehearse this off the live path; an export needing a source shutdown gets a separate downtime window.
- 04 Boot the VM on an isolated VLAN and verify VirtIO drivers, networking, DNS, clock and service startup. For either route, expose only service ports through the existing edge, verify TLS and health checks, and send synthetic requests. Check cloud access without metadata and compare error rates and latency percentiles with the cloud copy.
- 05 Ramp the weighted record: one per cent, ten, fifty, all of it. Hold each step a business day, check resolver answers with `dig` and measure the actual request split in service logs. Stop on the owner's error or latency threshold. DNS weight is approximate and cached clients will lag behind it.
- 06 Soak a week at full weight. Watch guest disks through Proxmox and application metrics, or node and pod health with `kubectl`. Exercise a service restart and check that configuration survives; scratch files must not become untracked application state.
- 07 Repeat across the stateless inventory, batched by owner. The reference fleet budgets twenty days of work across an eight-week cadence. Measure VM rebuild and export effort separately; do not squeeze stateful disk transfers into that allowance. Retain each cloud copy until its own soak and return-path checks pass.

ROLLBACK

Set the new copy's traffic weight to zero, wait for cached clients to drain, and confirm requests return to the cloud. Keep that copy healthy through the soak. There is no point of no return for this stateless path because its authoritative data never moved. If the new guest has accepted unique persistent writes, stop: returning requires reverse transfer or a restore that preserves those writes before traffic changes. A stale cloud disk or pre-cutover snapshot cannot supply them. That VM needed the separate stateful migration plan.

15 BUCKETS, CACHE AND QUEUES

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
MOVE	MANAGED OBJECT STORAGE, REDIS AND QUEUES	HIGH	30 DAYS

Copying bytes is the easy part. A live move also needs an ordered record of changes, retryable publishing and consumers that recognise the same job twice.

AWS	S3, ElastiCache and SQS	SQS now carries a megabyte a message, so a large payload may be sitting inline rather than as a pointer into a bucket.
GOOGLE	Cloud Storage, Memorystore and Pub/Sub	acknowledgements belong to a subscription; a second matching subscription receives its own copy, not a share of the first one's work.
AZURE	Blob Storage, Azure Cache for Redis and Service Bus	sessions and duplicate detection begin at the Standard tier; Basic has no topics.

BEFORE YOU START

ACCESS

- Read access to every bucket and the managed cache, and write on the Ceph gateway (Move 12)
- The queue inventory: ordering, visibility timeouts, dead-letter rules, retention and message size ceilings

SOFTWARE

- rclone 1.75.1, including rclone check --download where the stores share no usable checksum
- Valkey 8.0.11 with valkey-cli, and NATS JetStream 2.12.15 with the nats command line

PEOPLE

- The owner of each key pattern and each queue consumer: both switches are code changes

WHY THIS WORKS

Objects, disposable cache and queues can move while the application runs, but copying and changing endpoints alone cannot guarantee it. The zero-minute reference requires ordered object-change replay and durable queue publishing with shared deduplication. If the application cannot supply those, retain the service or rehearse a separate write-stop window. Sessions and locks remain state even when they expire. The owner must account for them before this Move is complete.

SWAP

An immutable bucket needs no change journal: copy it once and verify it before switching readers.

DO IT FASTER

Begin the bulk transfer as soon as the change journal is reliable; measure its duration from bytes and available bandwidth.

WATCH OUT

Signed links still target their original provider. S3 Signature Version 4 URLs last at most seven days, often less with temporary credentials; audit actual expiry and revoke old upload paths before handing off writers.

LEFTOVERS

Non-current object versions bill on after the live ones go, and cache nodes bill until the last client moves.

\$12,800/mo	\$1,150/mo	91%	0 min	8 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The cache nodes and managed queues, after all stateful key patterns have a home and the thirty-day return window ends; confirm their charges disappear on the next invoice. Any retained archive continues to bill and must leave before Move 20 closes its account.

NEEDS 12 Disks: what goes local, what goes on Ceph 14 The first service, end to end

THE RUNBOOK

- 01 Sort the object estate by size and last access. Compare retrieval and transfer costs with continued storage before retaining an archive. Inventory versions, retention locks, metadata and access policies; copying current objects does not migrate those contracts.
- 02 Before copying, capture every object change in a durable application journal, including overwrites and deletions with version ordering. Keep the source authoritative. Bulk-copy with `rc1one` to Ceph, then replay the journal in order; verify content with a common checksum or `rc1one check --download`. A multipart or encrypted object's entity tag is not a portable content checksum.
- 03 Test client listing, overwrites and multipart operations. Rehearse the writer handoff and reverse replay. Once caught up, route every writer through the journal to Ceph, then switch reads. Fence old writers, including signed uploads. Keep journalling through the return window; a final blind source-to-destination copy could overwrite new data.
- 04 Classify cache keys with their owner, regardless of expiry. Only values safely rebuilt from the authoritative database take a cold start on Valkey 8.0.11, checked with `valkey-cli`. Retain sessions, counters and locks until a separate state-transfer or expiry-and-reissue plan passes; Move 16's PostgreSQL replication does not move them.
- 05 Stand up NATS JetStream 2.12.15 beside the old queues. Use a durable outbox to retry each publish until both brokers acknowledge it, with one stable job identifier. Deduplicate effects in the shared authoritative datastore. Test ordering, payload limits, dead-letter handling and acknowledgement extensions on long jobs; broker-local deduplication cannot cover two brokers.
- 06 Switch producers to JetStream, finish old publish retries, and drain visible, delayed and in-flight work before stopping old consumers. Watch delivery with `nats`. Retain payloads and processing records independently for thirty days: SQS retention is at most fourteen days, and acknowledged messages are not a backup.

ROLLBACK

For objects, fence writers, restore new objects and replay ordered changes from the journal back to the source, verify them, then switch endpoints. Rebuild disposable cache from its authority; never re-enable a stale session or lock store. For queues, stop dispatch, restore unprocessed payloads from the retained outbox and reconcile processing records before resuming the old broker. An acknowledged job may already have changed the database or charged a customer. The point of no return for a configuration-only return is the first write or job processed only at the destination; thirty days of retained evidence supports recovery, not automatic reversal of effects.

16 POSTGRES, THE ONE THAT MATTERS

CUTOVER

15 MIN



LAYER	LEAVING	RISK	REVERSIBLE
MOVE	MANAGED POSTGRESQL	HIGH	7 DAYS

Fifteen minutes of stopped writes, at the end of a week in which the new cluster was caught up, backed up, and restored once on purpose.

AWS	RDS for PostgreSQL	the default parameter group cannot be edited, so <code>rds.logical_replication</code> needs a custom group and a reboot.
GOOGLE	Cloud SQL for PostgreSQL	<code>cloudsql.logical_decoding</code> requires a restart; retained write-ahead log can fill storage or trigger a billed increase when automatic growth is enabled.
AZURE	Database for PostgreSQL flexible server	<code>CREATE EXTENSION</code> fails until the name is in the <code>azure.extensions</code> allowlist, and extensions requiring <code>shared_preload_libraries</code> also need a server restart.

BEFORE YOU START

ACCESS

- Logical decoding enabled on the managed instance, with the reboot taken
- Credentials for a bucket on the object gateway from Move 12, separate from the cluster's
- The Kubernetes database platform from Moves 11 and 12, including physical-host replica separation

SOFTWARE

- CloudNativePG 1.30.0, with a pinned PostgreSQL image on the source's supported major version and a rehearsed failover
- Barman Cloud plugin 0.15.0 and the platform's cert-manager 1.21.1; the plugin requires CloudNativePG 1.26 or later
- `psql` and `pg_dump` no older than the managed server

PEOPLE

- The application owner, present for the window when writes stop

WHY THIS WORKS

The application already runs on your hardware from Move 14, in a VM or a container. This Move migrates its managed PostgreSQL service to CloudNativePG on Kubernetes, using Move 12's local NVMe and object gateway. The primary and standbys must occupy different physical hosts even when their Kubernetes nodes are guests. The initial copy runs before the maintenance window; a rehearsed restore and final replication catch-up make the fifteen-minute reference possible. A VM-only estate may keep its managed database under Move 04. An existing PostgreSQL VM does not need converting to CloudNativePG to leave the cloud: keep its VM deployment and use a separately rehearsed database replication, failover and backup plan at Move 18. The operator-specific steps below apply only when the inventory selects the Kubernetes database route.

SWAP

A reverse subscription can shorten rollback only if the managed service permits it and its replication loop prevention and sequence repair have been rehearsed. Keeping the old instance alone does not keep its data current.

DO IT FASTER

Tune `max_sync_workers_per_subscription` within the source's slot and destination's worker budgets. Initial table copies already support parallelism; splitting related tables across subscriptions loses their shared transaction boundary.

WATCH OUT

Replication carries rows only. Sequences, large objects, schema changes and tuned parameters stay behind, and a sequence left at one collides on the first insert.

LEFTOVERS

A disabled subscription still retains source WAL. Once rollback no longer needs it, remove the subscription and verify slot cleanup; normal subscription removal drops its slot, but a detached or orphaned slot needs separate cleanup.

\$19,500/mo	\$0/mo	100%	15 min	12 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The managed instance and read replicas after seven days, verified destination recovery and the required snapshot retention. Confirm the resulting charges disappear on the next invoice; an active instance cannot produce that evidence first.

NEEDS 12 Disks: what goes local, what goes on Ceph 13 Images, secrets and one-command deploys

THE RUNBOOK

- 01 List extensions, roles, large objects and table replica identities with `psql`. Recreate required roles and grants; provider-only extensions or unsupported objects block the Move until a tested alternative exists. Freeze schema changes and rehearse returning destination writes to the managed instance.
- 02 Apply `pg_dump --schema-only` to a CloudNativePG cluster with one primary and two standbys on separate hosts. Preserve primary keys and replica-identity indexes. Confirm every updated or deleted source table has a suitable replica identity; the schema dump does not copy cluster-wide roles.
- 03 Create the publication and subscription, keeping application writes off the destination. Monitor `pg_stat_subscription`, table synchronization and retained source WAL continuously. Require every table to finish its initial copy and application queries to pass before booking cutover.
- 04 Configure the Barman Cloud plugin's ObjectStore and cluster plugin settings for continuous archiving and scheduled full backups. Keep an independent off-site copy with its keys. Check WAL arrival and archive failures; backups confined to the rack do not survive its loss.
- 05 Restore from the off-site copy into isolation and replay to a recorded timestamp. Disable subscriptions and outbound jobs there; compare with checks recorded for that recovery point, not a changing production database. Measure recovery and confirm the recovered application works before allowing destination writes.
- 06 Open the window. Stop all source writers and wait for in-flight transactions. Record a final published transaction and confirm the subscriber applies it after every table is synchronized. Compare final data, transfer any separately handled objects and set destination sequences from source values. Abort to the source if checks exceed the rehearsed deadline.
- 07 Disable the forward subscription, fence source writers and switch every connection string. Start writes only on the destination, then verify VM and container clients, errors and query latency. Keep destination backups running and preserve the tested return procedure throughout the seven days.

ROLLBACK

Until destination writes resume, return connection strings to the source and release its write pause. The point of no return for that quick return is the first destination commit. After it, stop writers again and use the rehearsed reverse replication or restore-and-replay procedure to carry those commits back; verify data and sequence positions before reopening the source. A source backup predating cutover cannot recover newer writes. Retain the managed instance and its backups for seven days, and take destination backups throughout that period.

V RUN

The traffic, the pager and the last account. Every article about leaving the cloud stops before this stage, and every migration that regrets itself failed inside it.

4	3	3	10
MOVES	ZERO DOWNTIME	HIGH RISK	MINUTES, TOTAL

IF YOU ONLY DO THREE

17

THE FRONT DOOR

The new edge goes up beside the one carrying users, with its own address and certificates, proved from a host file ...

19

BACKUPS YOU HAVE RESTORED, AND THE PAGER

VM backups, cluster manifests and a continuous database archive, copied beyond the rack — then the drill that ...

18

GO-LIVE, AND HOW YOU ABORT

Shift traffic in measured steps, with both edges reaching the same data. Lower DNS lifetimes ahead of time and keep ...

THE FRONT DOOR · GO-LIVE, AND HOW YOU ABORT · BACKUPS YOU HAVE RESTORED, AND THE PAGER · CLOSING THE ACCOUNT

■ 1 MEDIUM RISK

■ 3 HIGH RISK

17 THE FRONT DOOR

CUTOVER

0 MIN



LAYER	LEAVING	RISK	REVERSIBLE
RUN	MANAGED LAYER-7 BALANCERS AND CERTIFICATE SERVICES	MEDIUM	IMMEDIATELY

The new edge goes up beside the one carrying users, with its own address and certificates, proved from a host file entry while real traffic goes elsewhere.

AWS	Application Load Balancer and ACM	DNS validation records must remain while the certificates need automatic renewal, including the old edge's return window.
GOOGLE	external Application Load Balancer and Certificate Manager	URL map, target proxy and forwarding rule are separate objects, and the reserved address outlives all three.
AZURE	Application Gateway and Key Vault certificates	the gateway retrieves certificates through a managed identity; losing access can disable the affected HTTPS listeners.

BEFORE YOU START

ACCESS

- The public address range from Move 08, with one address held back for the gateway
- Write access to the VM configuration repository or the Argo CD repository from Move 13
- DNS challenge credentials restricted to certificate validation records
- An export of every listener rule, redirect, rewrite and health check from the managed balancer

SOFTWARE

- For VMs: a supported reverse proxy and ACME client, plus Keepalived, with exact package versions and configuration pinned in the guest template
- For Kubernetes: `helm`, Envoy Gateway 1.8.4, and `cmctl` for the platform's cert-manager 1.21.1
- `openssl`, to inspect the certificate each listener actually serves

PEOPLE

- Whoever owns the two most valuable transactions, to walk them end to end first

WHY THIS WORKS

Build the edge beside the live one. Envoy Gateway routes to Kubernetes services; a pair of proxy guests routes to application VMs. Either uses an address from Move 08 and services tested on your hardware. Keep production DNS unchanged and restrict test access. Export the managed balancer's listener rules and rewrites into versioned proxy configuration or Gateway API objects, then prove their behaviour before shifting traffic. The figures retain the reference Kubernetes edge; price a VM proxy pair separately.

SWAP

For a small rule set, keep the mapping beside the route tests instead of in a separate table.

DO IT FASTER

Keep one low-traffic route as a permanent smoke test for the address, certificate and proxy.

WATCH OUT

Renewal does not guarantee reload. Alert on the certificate each proxy serves, and verify the standby's certificate before failover.

LEFTOVERS

Managed balancer and address charges continue until decommissioning. Keep certificate-validation records and permissions while the old edge remains a return path.

\$2,400/mo	\$320/mo	87%	0 min	4 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Nothing yet. The managed balancer, its listener rules and its certificate service can switch off after Move 18's retention window and a verified replacement.

NEEDS 14 The first service, end to end 16 Postgres, the one that matters

THE RUNBOOK

- 01 For VMs, deploy the pinned proxy template to guests on different physical hosts. For Kubernetes, install Envoy Gateway with `helm install eg oci://docker.io/envoyproxy/gateway-helm --version v1.8.4 -n envoy-gateway-system --create-namespace`. Configure at least two data-plane replicas on different hosts, verify the platform's cert-manager and commit the settings for Argo CD.
- 02 Assign one virtual IP from Move 08. For VM proxies, configure Keepalived to track proxy health and transfer it between guests on the same layer-2 network; permit its peer traffic and address announcements through the host bridges. For Kubernetes, use the cluster network's layer-2 announcement. Power off the physical host serving the address and measure recovery, then test a failed proxy process separately.
- 03 Use DNS validation before traffic moves. On VMs, schedule the ACME client and deploy renewed certificates to both proxies with a reload hook. On Kubernetes, issue through cert-manager and check `cmctl status certificate`. Rehearse renewal in staging, then inspect production with `openssl s_client`, supplying the destination address, SNI hostname and hostname-verification options. Repeat after failover.
- 04 Read the balancer export line by line into a table: one row per listener rule, path condition, redirect, header rewrite, sticky-session setting and health check. The rule nobody remembers is the one a customer depends on.
- 05 Translate each row into proxy configuration or Kubernetes HTTPRoute and any required policy objects, and write requests proving its observed behaviour. Enable backend health checks. Rule precedence differs between implementations: test overlapping matches so a previously shadowed rule cannot start serving unexpectedly.
- 06 Point a laptop at the new address with a host file entry and walk sign-in, the two most valuable transactions, a large upload and any websocket. Cut nothing over; Move 18 moves the traffic.

ROLLBACK

Nothing here serves users, so backing out is removing the test gateway or proxy guests and their address assignment, once no host file entry still points at them. There is no point of no return in this Move, by design. Keep the definitions in the repository so the edge can be restored and retested.

18 GO-LIVE, AND HOW YOU ABORT

CUTOVER

10 MIN



LAYER	LEAVING	RISK	REVERSIBLE
 RUN	WEIGHTED DNS AND THE PARALLEL MANAGED EDGE	 HIGH	7 DAYS

Shift traffic in measured steps, with both edges reaching the same data. Lower DNS lifetimes ahead of time and keep the old edge available for a week.

AWS	Route 53 weighted records	all-zero weights are treated equally; with health checks, a zero-weight endpoint can return if every positive-weight endpoint fails.
GOOGLE	Cloud DNS routing policies	a policy cannot coexist with a plain record of the same name and type, so creating one replaces that set.
AZURE	Azure DNS and Traffic Manager	endpoint weights run from 1 to 1000; remove an endpoint from rotation by disabling it, not by assigning zero.

BEFORE YOU START

ACCESS

- Write access to the authoritative zone, and a second person who has it too
- The change window agreed with whoever answers the support line
- For a stateful VM, a restored backup and a tested transfer and reverse-transfer plan approved by its data owner
- Restored off-site backups, external failure probes and a tested on-call escalation for every service about to take production traffic

SOFTWARE

- dig run from two networks you do not control, and a resolver you have never used
- argocd sync windows for Kubernetes, or the VM pipeline's deployment hold
- The application-native replication or backup tooling already tested for each stateful VM

PEOPLE

- One named decider, and the abort criteria they read out rather than argue

WHY THIS WORKS

The edge ramp changes where requests enter; it must not choose between competing copies of writable data. Stateless services already passed Move 14, and the reference database moved in Move 16. Both edges must reach the same authoritative workload and data before their weights can be interchangeable. Lower the time-to-live ahead of the day, check actual client behaviour, and let each soak span a traffic cycle. Stateful VMs deferred from Move 14 need a separate service window before this edge ramp. Seed their data, rehearse the final transfer and time the return path. The strip's ten minutes covers the reference edge change only. Record each VM's measured downtime separately; if a write stop exceeds an hour, split the migration by independently movable workload or retain that VM until a shorter method is proved. An image export does not acquire a zero-downtime guarantee by appearing earlier in the book.

SWAP

Where a contract forbids a freeze, run the ramp in the customer's quiet hours in smaller increments across a fortnight.

DO IT FASTER

Take the one per cent step a week early, on its own. It finds the certificate and header faults while there is time to fix them.

WATCH OUT

Some corporate resolvers and mobile stacks ignore the time-to-live outright and reach the old address for days. Hence the week.

LEFTOVERS

The lowered time-to-live is itself a leftover, multiplying query volume and query billing on a zone nobody reads.

\$8,900/mo	\$0/mo	100%	10 min	3 days	1 week
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The old managed edge, after seven days at full destination traffic and a confirmed return test. Replace weighted routing with a record for the destination before removing the policy. Egress charges fall only as cloud traffic ends; retained services and archives can still generate them.

NEEDS 17 The front door

THE RUNBOOK

- 01 Lower each relevant DNS lifetime to 60 seconds and wait at least its previous TTL before shifting traffic; allow a day where that is longer. Include Traffic Manager's profile TTL and any aliases or CNAME chain. Confirm with `dig` from independent networks.
- 02 Write the abort criteria as numbers: error rate above a figure, tail latency above a figure, any data-integrity report at all. Name the decider on it.
- 03 Tell the clients a record cannot move: mobile builds with a pinned host, batch jobs holding an old address, third parties allowlisting one.
- 04 For each deferred stateful VM, seed a destination guest while the source remains authoritative, using its tested application-native replication or backup tooling. Restore into isolation, verify record counts or file checksums, and rehearse final transfer and reverse transfer. Book its measured write-stop window separately from the edge change.
- 05 In that VM's window, stop all source writers and timers; keep destination writers stopped too. Drain replication or transfer the final delta, validate the data, then switch traffic and start only destination writers. Retain the source stopped. If verification misses the deadline before destination writes, abandon the cutover and resume the source. Never split live writes between independent VM disks.
- 06 Before the ramp, verify both edges reach the same authoritative service and data. Confirm an off-site restore has passed, backups continue, and an external probe failure reaches the named on-call engineer. The full estate drill in Move 19 does not defer these checks. An old edge unable to reach the authority cannot provide traffic-only rollback.
- 07 Open the edge window and freeze releases with an `argocd` sync window or the VM pipeline's deployment hold. A release landing mid-shift cannot be told apart from the shift.
- 08 Set weights for roughly one per cent, then ten, fifty and all traffic, soaking at each step. DNS weights select answers, not an exact share of requests: measure traffic, errors and latency. The ten minutes is the final reference step, excluding VM windows and soak time.
- 09 Keep the old edge live for seven days. On Azure, disable its Traffic Manager endpoint; elsewhere use zero weight with a positive survivor and verify health-check fallback behaviour. Probe its return route directly throughout the week. Nothing is deleted before that check passes.

ROLLBACK

For an edge failure, restore the old edge's traffic weight or re-enable its Traffic Manager endpoint, and confirm requests reach the same authoritative service; cached clients may take longer than one TTL. For a stateful VM, the point of no return for a quick switch back is its first destination write. Stop writers, reverse-transfer those changes or restore and replay them to the source, validate the recovered data, then restart source writers and change traffic. Never boot a stale source as a writable fallback. The managed edge stays available for seven days, but its existence cannot replace that data recovery procedure.

19 BACKUPS YOU HAVE RESTORED, AND THE PAGER

CUTOVER

0 MIN



LAYER RUN	LEAVING MANAGED BACKUP, ALARMS AND A RENTED PAGING SERVICE	RISK HIGH	REVERSIBLE IMMEDIATELY
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VM backups, cluster manifests and a continuous database archive, copied beyond the rack — then the drill that measures recovery time.

AWS	AWS Backup and CloudWatch alarms	after a compliance lock's grace period, retained recovery points cannot be removed; closing the account still deletes them after its closure period.
GOOGLE	Backup and DR Service and Cloud Monitoring	a backup vault's enforced retention makes a backup indelible until it expires, by you or by Google, so the copy outlives the decision to leave.
AZURE	Azure Backup and Azure Monitor alerts	enhanced soft delete can retain removed backups for 14 to 180 days; check the configured period before promising a vault closure date.

BEFORE YOU START

ACCESS

- Object storage with object lock at an independent provider
- Five engineer-days reserved for the rebuild drill

SOFTWARE

- `vzdump`, `qmrstore` and guest agents from Move 11's pinned templates
- `velero` 1.18.1 with its compatible storage plugin and Move 16's Barman Cloud plugin, pinned in Git
- `rc1one` 1.75.1 for the copy out to the third provider
- Prometheus and Loki, with versions pinned in Git and metric-series, log-stream and retention budgets agreed

PEOPLE

- The existing operations rota, with a second escalation contact
- Provider support contacts and approved physical-intervention runbooks

WHY THIS WORKS

The deliverable is the drill: rebuild from the repository and off-site backups while production stays untouched, with a stopwatch running. Compare measured recovery time with the agreed objective; a slow restore means the target is still unmet. Per-service backups and external paging already had to pass before cutover. This Move completes the estate-wide recovery exercise and operating handover. The cloud operations rota becomes the on-prem rota. Remote hands or the rental provider handles contracted physical faults within the facility or rental budget; your engineers own software recovery and incident command. The reference retains cloud staffing hours and salary, leaving potential savings from cheaper on-prem roles uncounted; validate actual costs in the pilot.

SWAP

If the drill on the spare machine is disruptive, rent a dedicated server by the month and rebuild into that instead.

DO IT FASTER

Automate the drill and record each run's duration. Repetition reduces manual work; data volume and restore bandwidth still set a floor.

WATCH OUT

A booted VM or healthy pod proves little about the data. Check volumes, database rows and application behaviour, and test a full host failure separately from the spare-machine restore.

LEFTOVERS

Backup vaults bill after the resources they protected are gone, and one under a retention lock stays until it expires.

\$4,800/mo	\$1,400/mo	71%	0 min	5 days	—
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF Managed backup schedules, alarms and rented pager seats, after destination recovery is timed and a test page reaches the on-call engineer. Retained vault contents and cross-region copies remain until their retention and audit obligations expire.

NEEDS 16 Postgres, the one that matters 18 Go-live, and how you abort

THE RUNBOOK

- 01 Confirm `vzdump` covers every required guest disk and VM configuration outside its datastore. Use guest agents for filesystem coordination and application-native backups for databases; save host configuration and passthrough data separately. Configure `velero` volume-data movement, not just CSI snapshots, plus Barman base backups and continuous WAL archiving. Set a private Velero repository password before its first backup.
- 02 Copy completed VM archives with `ncfone` to an independent provider. Carry Velero volume data and complete Barman backup-and-WAL chains off-site too. For mutable repositories, make a consistent checkpoint with writes and pruning paused before copying. Protect that copy with object lock and separate credentials; test retention against repository maintenance. Escrow decryption keys outside the cluster.
- 03 Temporarily activate a shelf spare or reserved rental server on an isolated network. Rebuild Proxmox and restore a representative VM with `qmrestore`, or rebuild a test Kubernetes cluster from Git and restore its volumes and database. Validate application data with outbound jobs disabled, then compare elapsed time and data loss against the recovery objectives. After the drill, return it to reserve and unplug an owned spare.
- 04 Check Prometheus metric-series limits and Loki log-stream and retention limits. Drop unwanted metrics before ingestion and avoid unbounded log labels. Include host quorum, backup and WAL-archive failures, storage health and free capacity, as well as application signals.
- 05 Put the alerting where the cluster cannot take it down. OneUptime runs the probes, the rota, the escalation and the status page from outside; the author of this book founded OneUptime, and the copyright page says so. Add a dead-man's-switch heartbeat.
- 06 Transition the existing cloud rota to the new platform. Route physical faults to remote hands or the rental provider under the support contract; keep software recovery with the operations team. Test both escalation paths, name a backup contact and record what waits until morning. Feed measured hours and staffing costs back into Move 03's budget.

ROLLBACK

Keep the last proven backup schedules and alert routes active until replacements pass a restore and a paging test. Managed backups of the old estate do not contain writes made on the new one. If a new schedule fails, repair it or restore the previous destination backup configuration without discarding existing copies. The point of no return for historical recovery is the expiry or destruction of its last usable copy or key. Retain provider backups for their required period and verify your independent recovery chain before removing any protection.



RUN · 0 MIN · HIGH RISK

Managed backup schedules, alarms and rented pager seats, after destination recovery is timed and a test page reaches the on-call engineer. Retained vault contents and cross-region copies remain until their retention and audit obligations expire.

20 CLOSING THE ACCOUNT

CUTOVER

0 MIN



LAYER RUN	LEAVING THE PROVIDER ORGANISATION, ITS AUDIT TRAIL AND ITS SUPPORT PLAN	RISK HIGH	REVERSIBLE NO
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The estate moved a month ago. The account did not: a dormant organisation still bills, still holds credentials, and still lets someone start an instance in it.

AWS	AWS Organizations and the management account	the management account can close members centrally, but a closed account's ninety-day reopening period does not cancel its commitments.
GOOGLE	organisation and its projects	project deletion is scheduled rather than immediate, and a lien refuses the request outright.
AZURE	subscriptions and the Entra tenant	tenant deletion also checks Microsoft 365 and other connected services; leaving Azure compute does not make their shared identity disposable.

BEFORE YOU START

ACCESS

- Provider closure authority and independent recovery logins, controlled by two named people
- The registrar login, often held by one person who has since left

SOFTWARE

- aws, gcloud or az signed in with an identity that can read billing
- rc1one 1.75.1 to copy exported audit trails and billing history somewhere you own

PEOPLE

- Whoever answers the auditor, because the attestations you inherited become yours to produce
- The finance owner who has to see the invoice reach zero and say so

WHY THIS WORKS

A migration that stops at cutover can leave live credentials, forgotten bills and an audit trail nobody has exported. Observe a full monthly job cycle before destructive cleanup: the job on the twenty-eighth still needs its bucket. Planned recovery copies keep billing during that wait. Close only after retention obligations, commitments and shared services are accounted for. The strip's zero is the completed reference outcome; an unexpired contract or retained archive still has a cost.

SWAP

Where one thing genuinely has to stay — an archive whose egress you will not pay — move it into a small account of its own and close everything around it.

DO IT FASTER

Inventory key dependencies during the observation month. Schedule destruction only after independent recovery passes: a key pending deletion can already be unavailable for decryption, even while deletion remains cancellable.

WATCH OUT

Committed spend does not end when usage does. A Savings Plans commitment keeps billing against the organisation after every resource inside it has gone.

LEFTOVERS

Check marketplace contracts, support cancellation terms and final invoices separately. Account closure is not proof that contractual charges or data-retention duties ended.

\$3,200/mo	\$0/mo	100%	0 min	4 days	1 month
WAS	NOW	SAVED	CUTOVER	EFFORT	WAIT

TURN OFF The unused account or project and its remaining services, after the observation month and verified export. Preserve required archives and shared identity, settle commitments, and reconcile the final invoices before calling its cost zero.

NEEDS 18 Go-live, and how you abort 19 Backups you have restored, and the pager

THE RUNBOOK

- 01 Take the final invoice apart with aws, gcLoud or az. Expect address translation gateways nothing routes through, CloudWatch log groups still ingesting, snapshots on a retention policy nobody reviews, a support plan priced against spend that has gone, and two subscriptions from a proof of concept three years ago.
- 02 Have the auditor confirm the required period, then export audit trails, billing history and compliance evidence through their provider services. Copy the exports with rcLone and test them without cloud credentials. Re-encrypt retained data under independently controlled keys where necessary; copying ciphertext does not free it from a provider-managed key.
- 03 Revisit the archive you decided in Move 15 to leave behind. Data parked in an account you are closing is not archived anywhere. Move it out, or agree in writing to delete it once someone has confirmed nothing still reads from it.
- 04 Transfer provider-managed domain registrations, preserving DNS service and renewal access, then restore the transfer lock. Move authoritative zones too before closing their projects or accounts; changing registrar alone does not move DNS hosting. Verify mail, certificate validation and the replacement platform's sign-in path.
- 05 Quiesce remaining workloads and disable automation that restarts them. Deallocate Azure VMs rather than only shutting down guests. Reduce or cancel support under its terms and observe one full monthly job cycle with no unexplained usage. Keep snapshots, volumes and their keys throughout the wait; finance records their expected residual charges.
- 06 With exports restored, retention satisfied and owners' approval recorded, remove remaining workload resources and unused credentials. Schedule key destruction only after proving no retained data needs them. Keep closure administrators until the last request. Close member accounts, projects and subscriptions before their parent, preserving any tenant still used for identity or office services.

ROLLBACK

During observation, retained resources and keys allow a tested return. Destructive cleanup is irreversible once required data or keys disappear: that is the point of no return, potentially before account closure. AWS offers a ninety-day reopening period subject to its conditions; Azure describes thirty to ninety days of data retention after cancellation. Google permits project restoration for thirty days, but some services can delete data sooner. None is a universal restore guarantee. Treat closure requests as final and rely on independent, already-restored copies for records you must retain.



RUN · 0 MIN · HIGH RISK

The unused account or project and its remaining services, after the observation month and verified export. Preserve required archives and shared identity, settle commitments, and reconcile the final invoices before calling its cost zero.

REHEARSAL RECORD

The first service produces evidence before it produces savings. Its database remains in the cloud, so the new variables are the host platform, network path and identity.

Move 14 · Why this works

Service / scope

Move / record reference

Date / time zone

Operator / reviewer

Starting state

Identify every disk write, scheduled job and metadata dependency. Disable duplicate jobs in the test copy, and keep it isolated from production traffic until the owner confirms it cannot create a second writer.

Move 14 · Runbook step 1

Service checks

For either route, expose only service ports through the existing edge, verify TLS and health checks, and send synthetic requests. Check cloud access without metadata and compare error rates and latency percentiles with the cloud copy.

Move 14 · Runbook step 4

Stop conditions

Write the abort criteria as numbers: error rate above a figure, tail latency above a figure, any data-integrity report at all. Name the decider on it.

Move 18 · Runbook step 2

Measured recovery

Validate application data with outbound jobs disabled, then compare elapsed time and data loss against the recovery objectives.

Move 19 · Runbook step 3

Result and evidence

A booted VM or healthy pod proves little about the data. Check volumes, database rows and application behaviour, and test a full host failure separately from the spare-machine restore.

Move 19 · Operator's notes: Watch out

Runbook excerpts and space for observations.

RESTORE PROOF

The deliverable is the drill: rebuild from the repository and off-site backups while production stays untouched, with a stopwatch running. Compare measured recovery time with the agreed objective; a slow restore means the target is still unmet.

Move 19 · Why this works

Service / scope

Move / record reference

Date / time zone

Operator / reviewer

PostgreSQL recovery copy

Restore from the off-site copy into isolation and replay to a recorded timestamp.

Move 16 · Runbook step 5

Protection and keys

Protect that copy with object lock and separate credentials; test retention against repository maintenance. Escrow decryption keys outside the cluster.

Move 19 · Runbook step 2

PostgreSQL isolation and comparison

Disable subscriptions and outbound jobs there; compare with checks recorded for that recovery point, not a changing production database.

Move 16 · Runbook step 5

PostgreSQL application and elapsed time

Measure recovery and confirm the recovered application works before allowing destination writes.

Move 16 · Runbook step 5

Next drill

Automate the drill and record each run's duration. Repetition reduces manual work; data volume and restore bandwidth still set a floor.

Move 19 · Operator's notes: Do it faster

Runbook excerpts and space for observations.

CUTOVER DECISION

The edge ramp changes where requests enter; it must not choose between competing copies of writable data.

Move 18 · Why this works

Service / scope

Move / record reference

Date / time zone

Operator / reviewer

Proceed / hold / abort

Write the abort criteria as numbers: error rate above a figure, tail latency above a figure, any data-integrity report at all. Name the decider on it.

Move 18 · Runbook step 2

Readiness evidence

Before the ramp, verify both edges reach the same authoritative service and data. Confirm an off-site restore has passed, backups continue, and an external probe failure reaches the named on-call engineer.

Move 18 · Runbook step 6

PostgreSQL write authority

Disable the forward subscription, fence source writers and switch every connection string. Start writes only on the destination, then verify VM and container clients, errors and query latency.

Move 16 · Runbook step 7

PostgreSQL point of no return

The point of no return for that quick return is the first destination commit. After it, stop writers again and use the rehearsed reverse replication or restore-and-replay procedure to carry those commits back; verify data and sequence positions before reopening the source.

Move 16 · Rollback

Old edge return window

Keep the old edge live for seven days. On Azure, disable its Traffic Manager endpoint; elsewhere use zero weight with a positive survivor and verify health-check fallback behaviour. Probe its return route directly throughout the week. Nothing is deleted before that check passes.

Move 18 · Runbook step 9

Runbook excerpts and space for observations.

OPERATING REVIEW

Feed measured hours and staffing costs back into Move 03's budget.

Move 19 · Runbook step 6

Service / scope

Move / record reference

Date / time zone

Operator / reviewer

Invoices and period

Fill the cloud column from the three invoices at the rate you actually paid, discounts in and tax out across all columns. One monthly number, checkable against Move 01.

Move 03 · Runbook step 2

Retained services

Add the services that stay rented. Move 04 identifies the edge, outbound mail and scrubbing; later Moves retain archived object storage, a registry, a queue and an off-site backup copy. Their cost is already in the cloud invoice and must remain in both metal columns.

Move 03 · Runbook step 5

Hours and migration costs

Validate recurring hours, skills and on-call cover in the pilot, then budget measured differences. Charge setup, training and migration once.

Move 03 · Runbook step 4

Health and capacity

Include host quorum, backup and WAL-archive failures, storage health and free capacity, as well as application signals.

Move 19 · Runbook step 4

Escalation and handover

Route physical faults to remote hands or the rental provider under the support contract; keep software recovery with the operations team. Test both escalation paths, name a backup contact and record what waits until morning.

Move 19 · Runbook step 6

Runbook excerpts and space for observations.

A FEW THINGS WORTH KNOWING

None of these is a Move. They are the 8 things that make the other 20 work, and most of them are the things people learn in the second year rather than the first.

THE BILL LAGS THE CHANGE BY A MONTH

A resource stopped on the third still bills for the first two days, and a reservation you no longer use can bill until its term ends. Reconcile the later invoices, not only the console.

TWO PEOPLE KNOW, OR NOBODY DOES

A platform that only one engineer can recover has a staffing failure domain. Have the second operator follow the written recovery procedure without prompting.

KEEP ONE FOOT IN THE CLOUD ON PURPOSE

An account with a working credential, a Terraform state and a warm standby is not a failure of nerve. Price that recovery option against its restore time and ongoing support, and retain the account deliberately if it meets the requirement.

WRITE THE RUNBOOK BEFORE YOU NEED IT

Every Move in this book is a runbook because that is the artefact that survives the person who wrote it. A migration that leaves no runbooks behind has to be done again by whoever comes next.

CAPACITY YOU OWN IS CAPACITY YOU HAVE

The cloud taught a generation to treat headroom as waste. On your own metal, idle capacity absorbs failures and growth. Price the measured peak, platform reserves and loss of the largest host, as Moves 05 and 06 require.

PLAN FOR HARDWARE FAILURES

Track disk, fan, PSU and optic failures against your actual fleet. Keep spares, response terms and recovery drills; an average failure rate does not predict which component fails next.

MEASURE THE SAVING AFTER THE MOVE

Reconcile the new invoices and recorded operating hours against the original decision. Control over the platform is useful, but it does not prove a saving.

YOU CAN GO BACK

The industry talks about repatriation as though it were one-way. It is not. 1 of the 20 Moves here are labelled irreversible. The other 19 have conditions on return; read those before starting.

A useful infrastructure decision may be unglamorous: a smaller bill, a restore that works, or a service that is easier to run. Record what changed and measure the result.

THE BOOK

20 Moves across 5 stages. 18 plan for zero downtime and 1 cannot be undone. Reference cutovers total 25 minutes; rehearse each against your own estate.

THE TYPE

Set in Archivo, drawn by Omnibus-Type, and JetBrains Mono, drawn by Philipp Nurullin and Konstantin Bulenkov. Both are open source. Every number in this book is set in the mono.

THE NUMBERS

Prices combine dated cloud list rates and illustrative hardware, facility and staffing allowances. Savings are model results, not quotations. Check the arithmetic against your own invoice.

Disclosure: the author founded OneUptime, which this book recommends for alerting, on-call, incidents and status pages. It is recommended because it is Apache-2.0 licensed, self-hostable, and does in one place what that Move would otherwise ask you to assemble from four separate tools. The alternatives named beside it are real ones, and a reader who prefers them loses nothing else in this book.